GEK-65411

DRIVE SYSTEM INSTRUCTIONS

MAINTENANCE MANUAL
For
CONTROL SIGNAL HIGHWAY
Driven By
Programmable Master Controller

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

MAINTENANCE MANUAL

FOR

CONTROL SIGNAL HIGHWAY

Driven By

Programmaole Master Controller

(GEK-65411)

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The information contained herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, USA.

HOW TO USE THIS MANUAL

This manual is a user's guide to the maintenance of General Electric's Control Signal Highway which is driven by a Programmable Master Controller.

We have written this manual to guide user maintenance personnel in the installation, troubleshooting and maintenance of a Control Signal Highway.

We recommend that the inexperienced user read the book from cover-to-cover to acquire an overview of the equipment maintenance requirements. Thereafter, the book will be a useful reference manual and should be kept near the equipment.

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--SECTION 1--
INTRODUCTION TO THE CONTROL SIGNAL HIGHWAY

GENERAL INFORMATION

This manual is a guide to the installation, operation, maintenance, and troubleshooting of General Electric's Control Signal Highway (CSH) equipment. Section 1 describes CSH system structure, operating principals, application rules and message structure. Section 2 describes master and slave stations and how they operate. Section 3 includes important installation guidelines and installation checkout procedures for a newly installed CSH system. Section 4 describes to you ,the operator, how a CSH system functions and how it is controlled. Troubleshooting procedures are aided by use of a utility program called CIAD, whose operation is described in Section 6.

We assume that a user of this manual will have some knowledge and experience in electrical and electronics technology. Some prior experience with industrial control systems is also helpful.

RELATED DOCUMENTS

Control Block Language Reference Manual GEK-65426
Field Support System Operator Manual GEK-65435
Drive and Direct Control Equipment GEK-61680A
Hewlett-Packard 21MX E-Series Computer Installation and Service Manual
Operating Manual for Programmable Master Controller and Installation Maintenance Module GEK-65442
Highway Test Module User Manual GEK-65151

CSH ARCHITECTURE

The CSH is structured in the form of a transceive loop. As shown in Figure 1-1, a master station is located at the head of the loop which contains from one to 15 slave stations.

The master station formats and transmits messages around the transceive loop. Signals returning from around the loop are received by the master station which checks the messages for errors and strips the variable information, (data) from the message for use by the Programmable Master Controller (PMC).

Each slave station around the loop is addressable and contains a repeater which restores and re-times the message signal at the repeater output. This output becomes a delayed replica of the output of the master station at the head of the loop. The purpose of the slave station is to interpret the messages as they pass through. Thereafter, each slave station acts upon the messages addressed to it, ensuring that the control information is received and acted upon at its destination.

The function of the PMC is to supply the content for the message format in prescribed bursts such that coordinated control of all devices connected to the slave station(s) is possible. Overall control of the PMC is maintained by the Hewlett Packard 21MX E-Series computer.

There is an optional provision for a redundant loop. (See Figure 1-2). If the primary loop is broken, the system automatically selects the redundant loop. The system will continue to operate on the redundant loop even when the primary loop is restored. You can cause the system to return to primary loop operation by re-initializing the system. This is accomplished by momentarily switching the MODE SELECTION switch (on PMC console) to OFF LINE, then back to ON LINE. If the primary loop is broken, it should be restored as soon as possible. If the redundant loop becomes broken while being used, the system will attempt to operate on the primary loop again, and it will do so provided the primary loop has been restored. However, if both the primary and redundant loops are open, the system will automatically switch back and forth between them for approximately 20 milliseconds while testing for a complete circuit.

Many loop configurations are possible to accommodate a wide range of application requirements. (See Figure 1-3). A DS3820 MMP master station can be used on one, two, or three loops. Its role on any given loop may be either that of a master or slave station. When used as a slave station, the capability of transferring message data in either direction at the PMC exists. In Figure 1-3, the DS3820 MMP-II heads loops II-1 and II-2. Master station DS3800 MMP-I heads loops I-1 and I-2. Note, however, that DS3800 MMP-I is also part of loop II-2 where it assumes the role of a slave. There can be only one master station on a loop. In the system of Figure 1-3, loop II, the master station is DS3820 MMP-II. The DS3820 MMP on each loop can be station configured either as a master station or a slave station on a per requisition basis by means of jumpers. (Jumpering is described in Section 3).

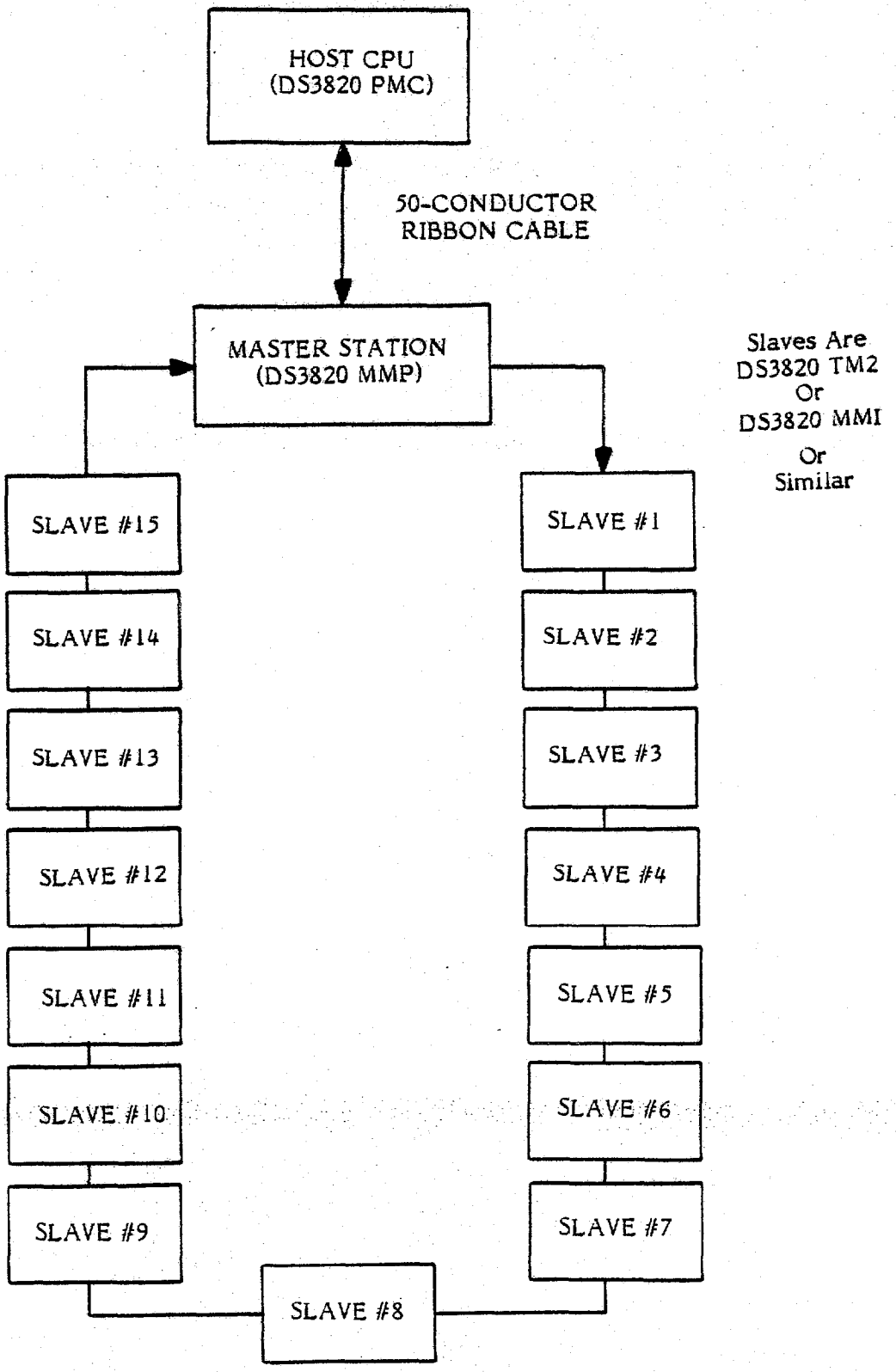


Figure 1-1.
Basic CSH Loop

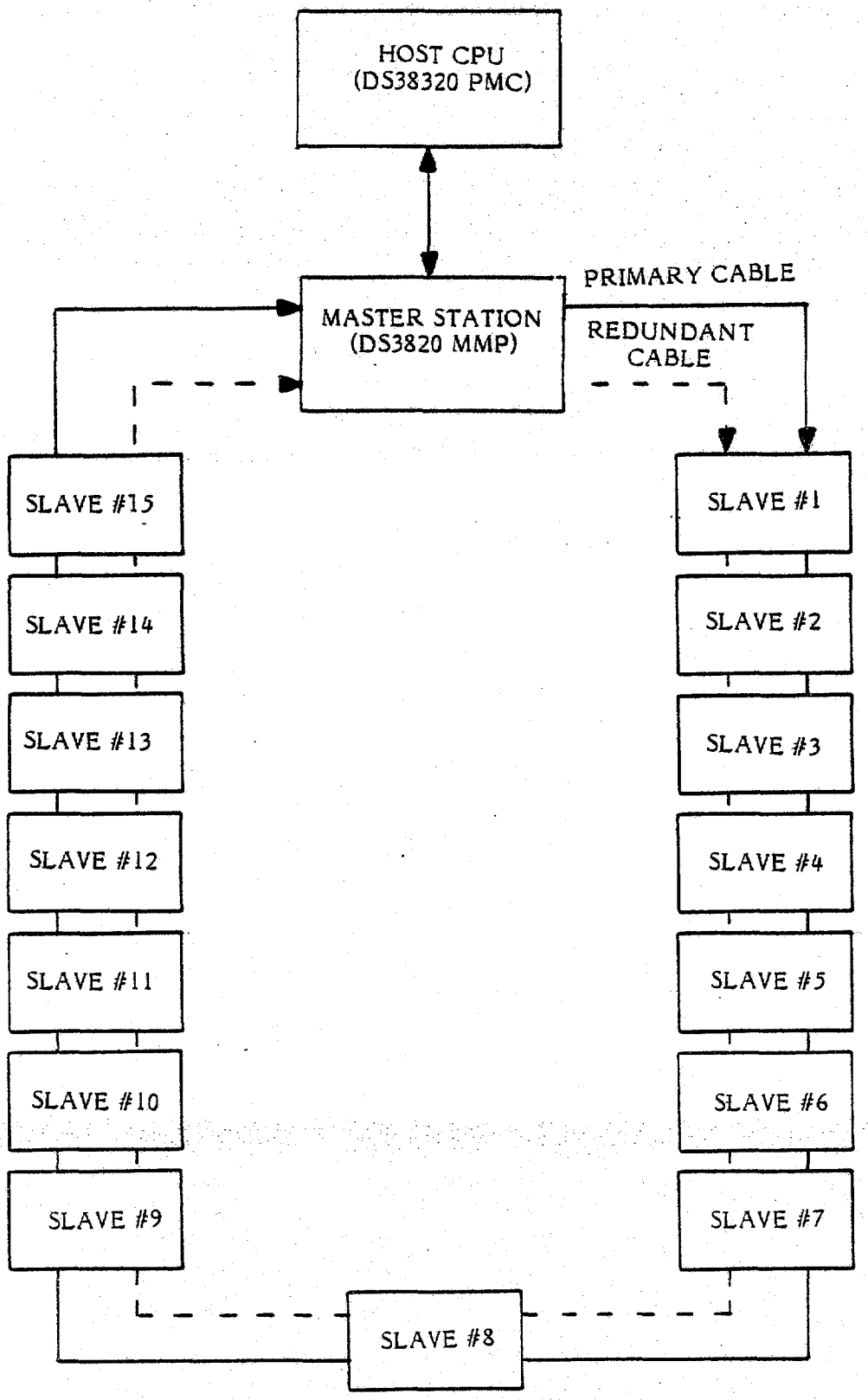


Figure 1-2.
Redundant Cable Option

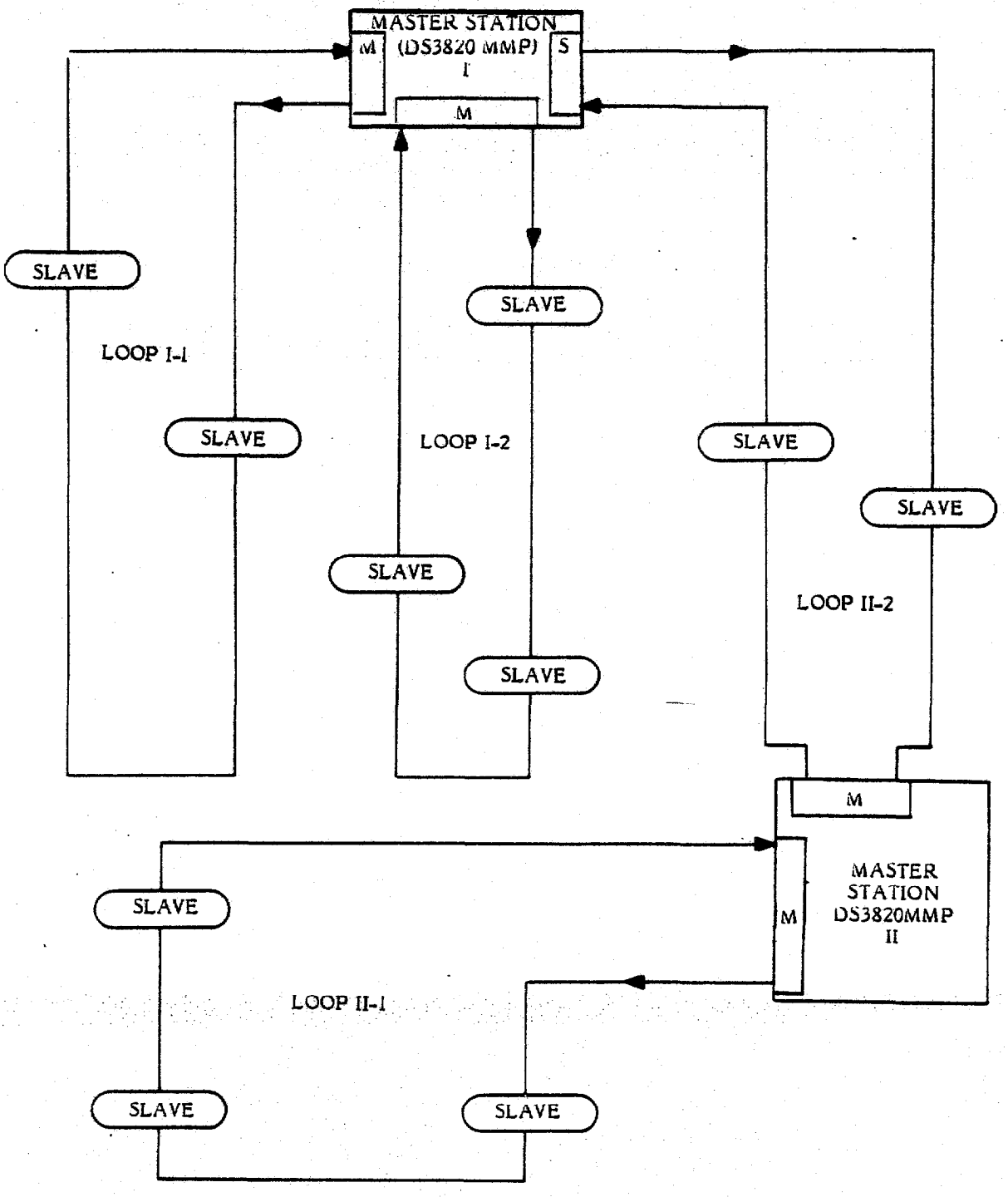


Figure 1-3.
Possible Loop Configurations

A typical process automation computer (PAC)-headed loop is shown in Figure 1-4. Such a loop maintains communication with all PMCs under the supervision of the process automation computer. On such a PAC loop, there are one or more operator station(s) to allow an operator to intervene in the operation of the PAC or any of the subordinate PMCs. Therefore, such an operator's station is depicted as being served by the PAC loop of Figure 1-4.

CSH APPLICATION RULES AND GUIDELINES

1. Up to 15 addressable slave stations may exist along a CSH loop.

2. The strappable station addresses of all slave stations along a CSH loop must be unique and cannot be "00" Hex. (See Section 6 for strapping information).

3. Multiple DS3820 MMP modules can be installed along a CSH loop. However, all but one must be strapped to slave mode. The slaves must then be station address-strapped per rule 2 above.

4. The maximum permissible loop length, L_{max}, in units of 1000 cable-feet is expressed by the following mathematical relationship:

$$L_{max} = 14.27 - 0.42S$$

"S" (above) is the number of slave stations along one loop.

5. The cable forming the CSH loop is Belden 8213 coaxial cable terminating at its ends in UG21BU connectors.

6. The Belden 8213 cable transitions to RG179 cable at each repeater through a coaxial connector plate provided in the PMC and slave cabinetry. The RG179 cable is provided as part of the equipment. The 8213 cable is provided by the customer.

7. The maximum recommended cable distance connecting three consecutive repeaters is 5000 feet.

8. The operating serial bit rate is 1.152 megahertz.

9. The bit error rate is less than 10^{-9}. The bit error rate is computed as following:

$$\text{Bit Error Rate}_2 \text{ (Error Bits/Sec}^2\text{)} = \frac{\text{Errors} \times 1.152 \times 10^6}{\text{Seconds Between Errors}}$$

10. The CSH loop is electrically isolated at each repeater and will reject up to 8 volts peak-to-peak common mode out to 10 megahertz.

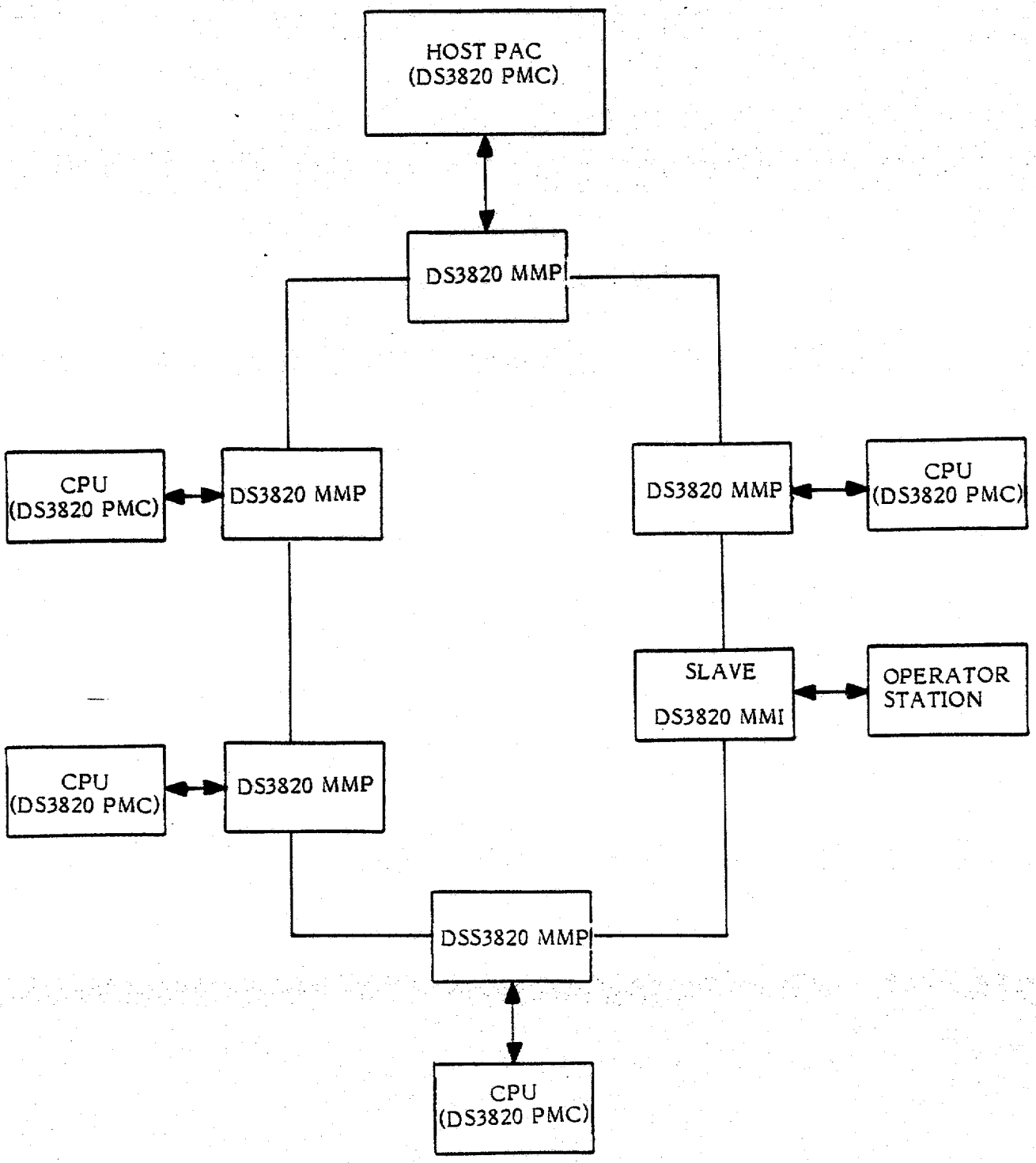


Figure 1-4.
The Process Automation Computer.

11. The isolators, cable connectors, and the 8213 cable are all rated to withstand 500 VDC, minimum.

12. CSH loops will operate reliably cross-temperature from 0 to 50°C. This means that one station along the loop can be in a 50°C environment while another station along the same loop can be in a 0°C environment.

13. Each CSH loop is grounded at a single point to the shield of the coaxial cables. The single point is at the head of the loop. This prevents error degradation from signals generated by ground loops. The ground wire is provided at the factory and connects the appropriate coaxial connector plate to the cabinet frame. The cabinet frame must be grounded to earth by the customer at installation time.

PMC CONTROL PANEL CONTROLS AND INDICATORS

The CSH is controlled by software within the PMC. User interface to that software is via the PMC control panel. (See Figure 1-5)

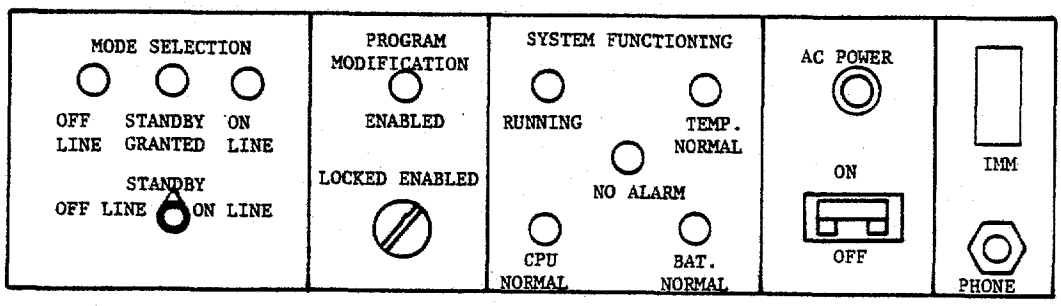


Figure 1-5.
PMC Control Panel Layout

PMC CONTROL PANEL CONTROL OR INDICATOR	FUNCTION OF PMC CONTROL
MODE SELECTION REQUESTED Switch	Controls CSH driver software.
OFF LINE	CSH driver software is disabled resulting in CSH stall.
STANDBY	Allows only messages that accept information from the controlled process.
ON LINE	Allows communication with the controlled process via the slaves in both directions.
MODE SELECTION Lights(Amber).	OFF LINE, STANDBY, and ON LINE lights come on to indicate that the mode selected by the REQUESTED switch is in operation.

SYSTEM FUNCTIONING
Lights (Amber)

NO ALARM	Remains lit as long as no uncorrectable errors have been detected within the CSH messages. The NO ALARM light is relightable by pushing on the indicator. If this light goes out while the system software is functioning, indicator lights on the front panel of the PMC computer will display the condition that caused the NO ALARM light to go out. This is reliable only during system operation.
RUNNING	Remains lit as long as the PMC program is not stalled.
CPU NORMAL	Indicates proper operation of the CPU power supplies.
TEMP NORMAL	Monitors PMC cabinet temperature. If this light goes out, it indicates that the PMC cabinet is too hot and the system should be safely shut down.
BAT NORMAL	Not Applicable

PROGRAM MODIFICATION

ENABLED Light	Comes on when the software recognizes the "ON" position of the key switch.
Key Switch	LOCKED Position - prevents a change of software status from the Intermediate Maintenance Module (IMM). ENABLED Position - allows the IMM operator to make a software change.
AC POWER	
ON - OFF Switch	Controls AC power to the DS3820 MMP.
PHONE Jack	Allows the operator to communicate with down-line personnel.
IMM Jack	Accommodates the IMM terminal signal cable.

DS3820 PMC CABINET

The Hewlett-Packard computer, PMC Control Panel, DS3820 MMP master station boards and other associated equipment are contained in the PMC cabinet. Refer to Figure 1-6 for the location and appearance of this equipment.

Physical Description

PMC Cabinet Weight: 600 pounds (272 Kg)

Dimensions: Width - 24 inches(61 cm)
Depth - 36 inches(91.4 cm)
Height- 90 inches(228.6 cm)

Cooling: The PMC panel is cooled by two fans. The fans are mounted at the top of the panel below a screened opening. Air enters the cabinet through a filter in the front door port.

Overtemp: A thermostat is located below the power supply. If the temperature approaches an unsafe operating level, the overtemperature indicator will activate. If the temperature exceeds the operating level, power to the power supply is automatically disconnected.

Electrical Description

Input Power Requirements:

87 to 127 volts rms. measured between input terminals ACLH and ACLL. (Located between TB2-2 and TDB2-1).

Power Supply PS5 Outputs:

OUTPUT	POWER SUPPLY CONNECTORS	PIN(S)
+5 VDC, 40 Amps	JA thru JF	5,6
	JA thru JF	6,7
+15 VDC, 2 Amps	JA thru JF	10
-15 VDC, 2 Amps	JA thru JF	11,12
+28 VDC, 5 Amps	JA thru JF	8
Logic 1	JA thru JF	1
Logic 0	JA thru JF	2

CSH PROTOCOL

Messages are contiguous and are identified from each other by a lack of modulation for a 1 1/2-bit period at the beginning of each message. (See Figure 1-7). The modulation is bi-polar and phase modulated as shown in Figure 1-7. Each bit time contains a pair of opposite-polarity pulses. The order of the pulse pair determines whether a binary one or zero is encoded. The encoded information is shown (in terms of message fields) just below the modulation sketch in Figure 1-7.

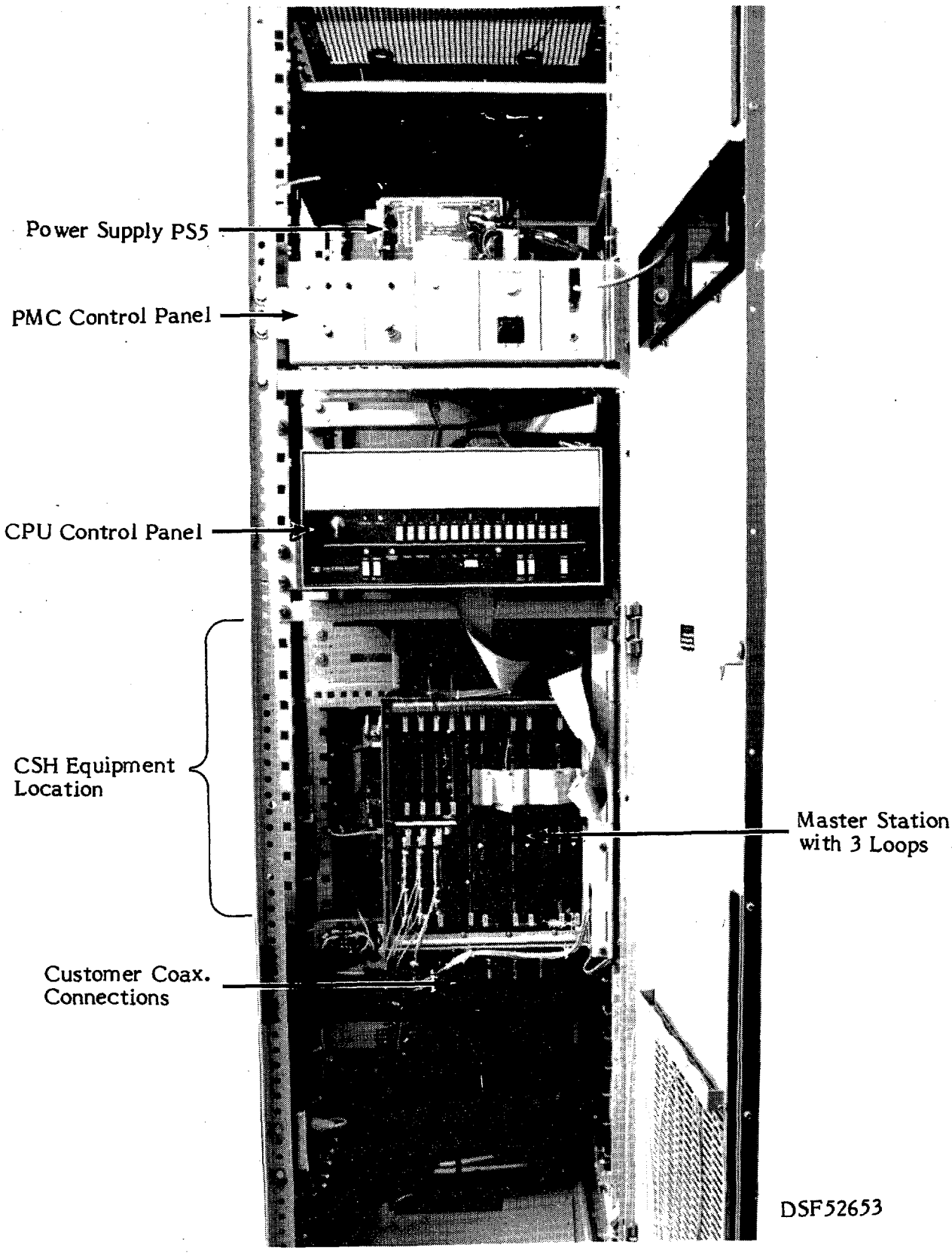


Figure 1-6.
DS3840 PMC Cabinet

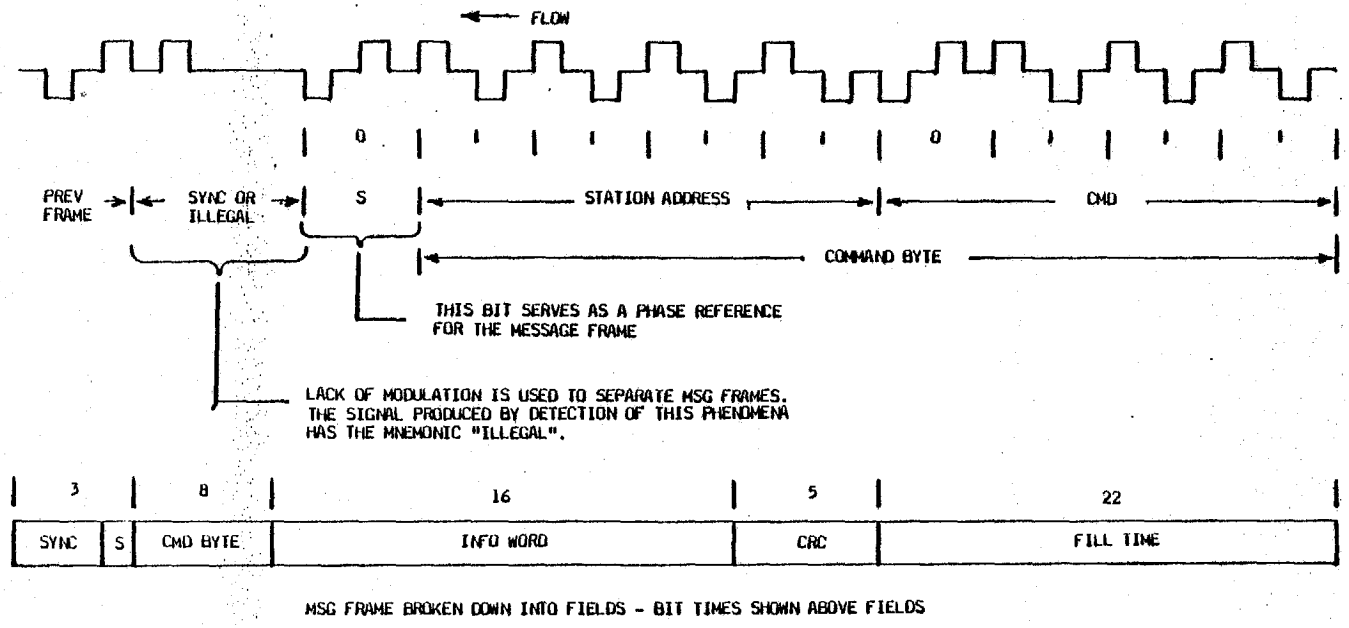


Figure 1-7
CSH Message Format

The SYNC field is the message separator. The "S" within the sync field is a single zero bit that allows phase lock onto the message.

The CMD BYTE contains the addressee of the message (i.e., CSH slave station) in 4 bits followed by the intent of the message in the next 4 bits. The intent of the various message types is described in Table 1-1.

Table 1-1
CSH Commands

MNEMONIC	CMD BYTE CODE	MESSAGE INTENT
IDLE	00	IDLE: Occurs whenever there are no other messages to be transmitted
SYNC	X0	SYNCHRONIZE: Message searches for the presence of station "X". The purpose is to establish loop viability before waking up the PMC for CSH management. Message occurs during power-up or after link errors.
ACCA	XA	ACCEPT ADDRESS: Transmits 8 bits of device address information to awake a device served by station "X" for the purpose of sending it control information in the next ACCD message.
ACCD	X9	ACCEPT DATA: Causes information to be sent to a previously awakened device served by station "X". After supplying this information to the served device, the slave station automatically increments the device address.
TLKA	XB	TALK ADDRESS: Transmits 8 bits of device address information to awaken a device served by station "X" for purposes of acquiring information during the next "TLKD" message.
TLKD	XC	TALK DATA: Causes information to be acquired from a previously selected device served by station "X". After acquisition of this information from the selected device, the slave station automatically increments address to the next sequential device address.
INQ	XE	INQUIRY : This message is sent to station X to query it for the nature of its demand for special attention during a previous IDLE message.
INQR	XF	INQUIRE AND RESET: Same as INQ except that the reset line to I/O devices is activated.
ACK	X2	ACKNOWLEDGE: An acknowledgement to the slave station X that its needs for special attention will be honored.
ACCT	X8	ACCEPT TEST DATA: This message deposits test data within slave station X.
TLKT	XD	TALK TEST DATA: This message retrieves the test data from station X.

The INFO WORD field carries the variable information within the message. Its contents can be addresses of devices served by the slave station. The contents can also be information to or from devices served by a slave station. When INFO WORD carries information from served devices, the field is encoded as all zeros by the master station. Then this field is appropriately inverted by the addressed slave station according to information supplied to that slave station by the selected, controlled device.

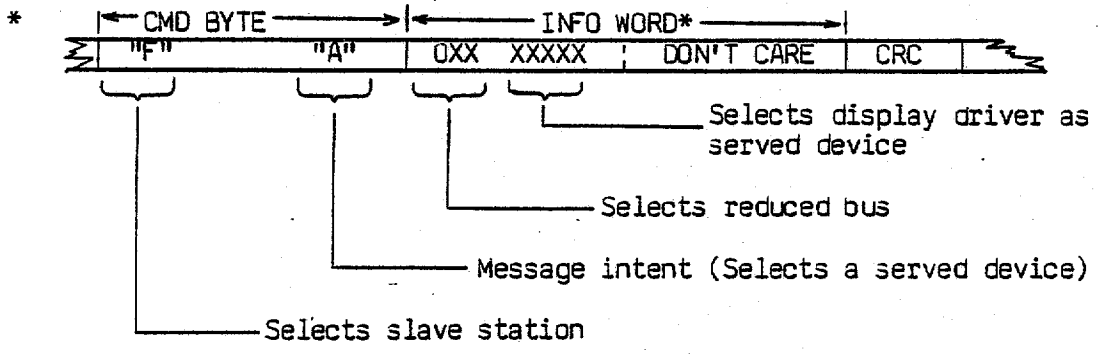
The extraction or insertion of information within the INFO WORD field occurs on-the-fly as the message passes through the repeater.

The CRC field checks the message content for errors over the span of the CMD BYTE field plus the INFO WORD field.

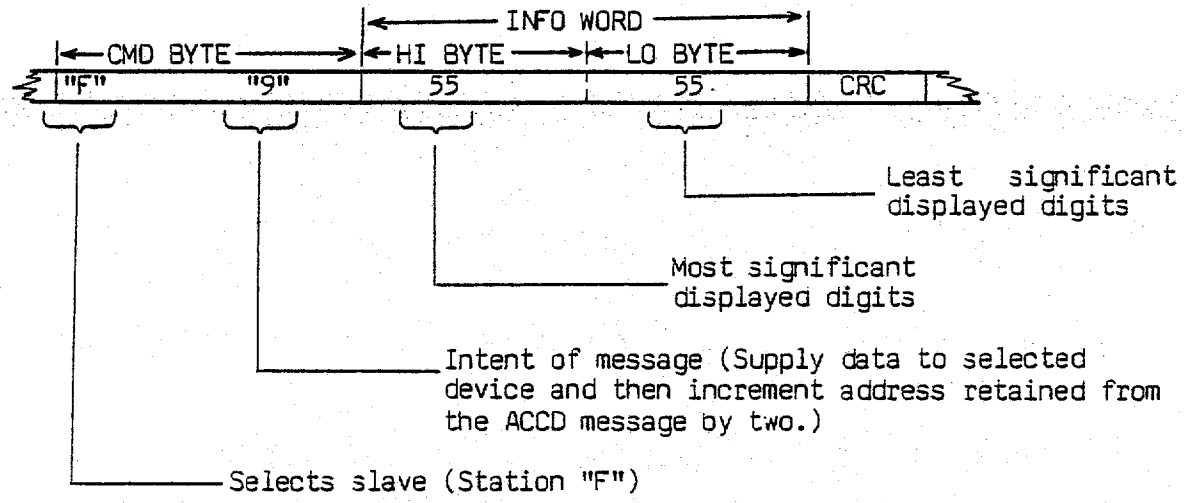
The FILL field has no other function than to absorb time at least equal to the longest permitted propagation delay around a CSH loop.

Information Transfer via CSH

During the following example of information transfer via CSH, refer to Figure 1-8. Consider the case of two types of messages listed in Table 1-1 which are sent to a remote device. They are an ACCA message and an ACCD message. The two messages are encoded and formatted as diagrammed below:



* X denotes a variable binary bit



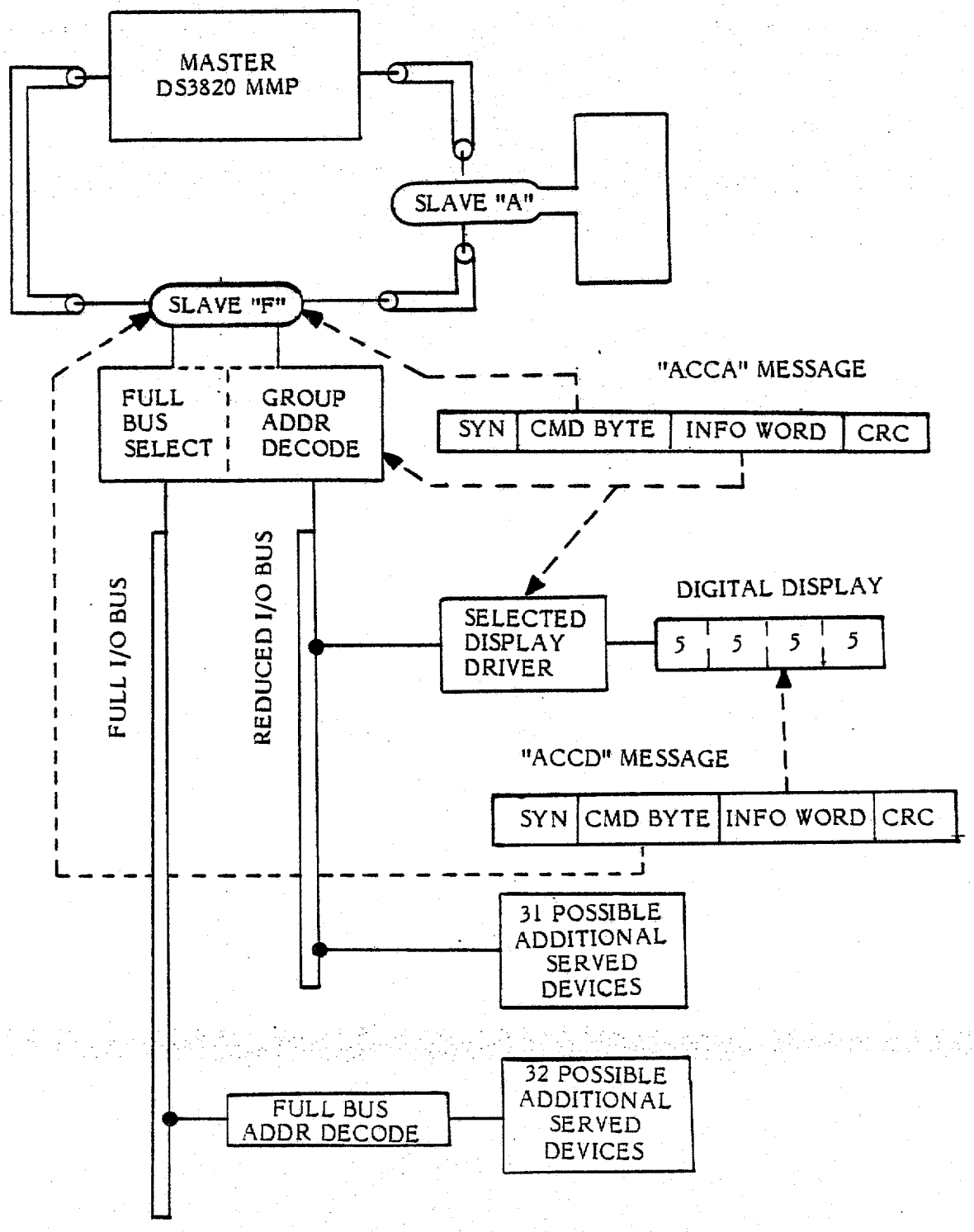


Figure 1-8.
CSH Information Transfer

Some devices (not illustrated) will accept only two digits of information (8-bits) although the message supplies 16 bits of information. Those devices are hard-wired to accept the information in the least significant byte and ignore the information in the most significant byte.

There are also data-supplying devices (not illustrated) capable of supplying either 8 or 16 bits of information into the INFO WORD field of a "TLKD" message. In this case, the eight bit device would supply the same information into the high byte and the low byte of INFO WORD field.

During actual operation, messages are grouped into segments of data output messages and segments of data-gathering messages. The various message segments are run at fixed or variable time ticks as required by the application.

The CSH has the capability of accepting slave demands encoded into IDLE messages which require the manual intervention of the operator. Each station address has a dedicated bit assigned to it within the INFO WORD field of an IDLE message. This allows a demand to be asserted for special attention in these exceptional cases. The most significant bit is dedicated to station "F"; the next bit of lesser significance to station "E" and so forth.

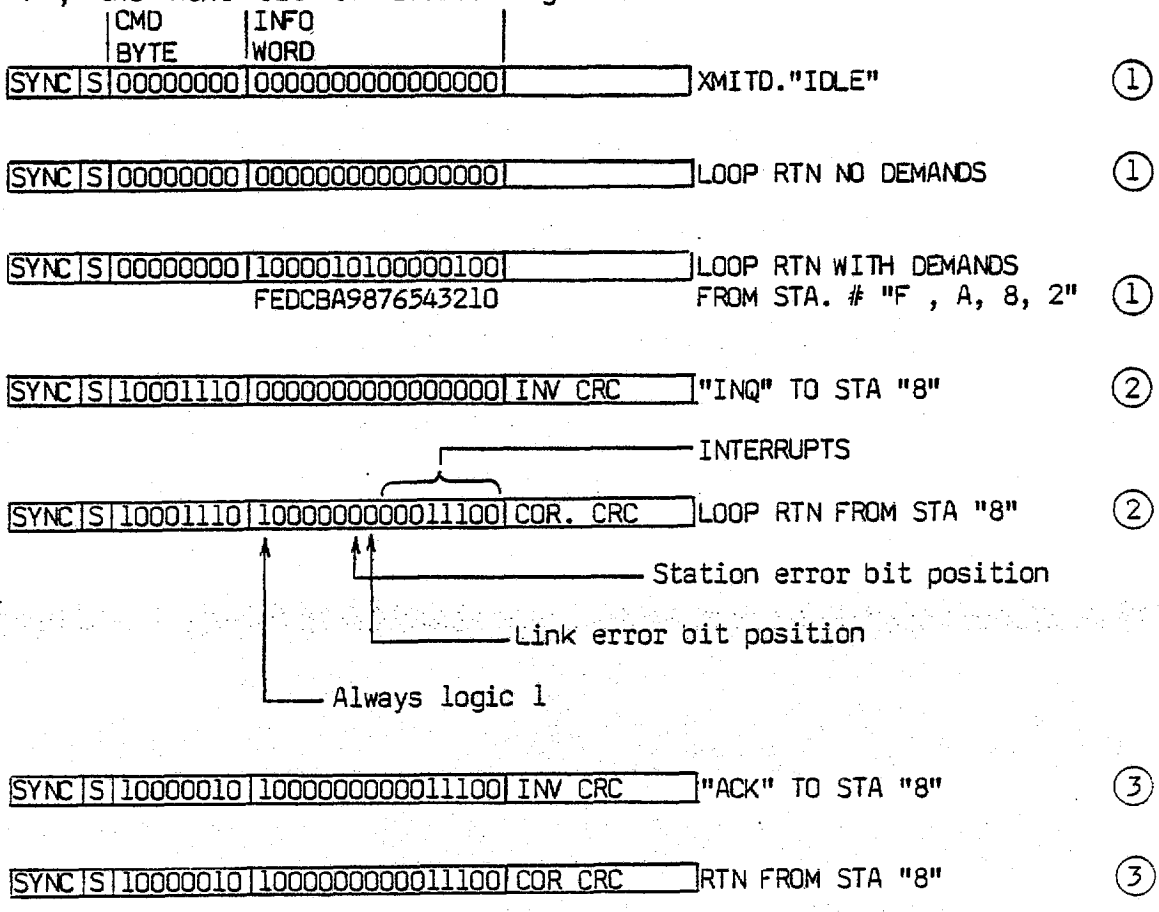


Figure 1-9.
Remote Interrupt Servicing Messages

Figure 1-9 depicts a sequence of messages that accomplish the following:

(1) Demands are discovered from multiple slave stations.

(2) A query is sent to one of the slave stations to determine the nature of its demand.

(3) An acknowledgement of the receipt of the query response is sent back to the slave station. This is required to turn off demands.

The encircled numbers above are a cross reference to the actual messages in Figure 1-9.

Referring to message 2 of Figure 1-9, the LNK error bit, had it been present, would have denoted the detection of a faulty message during the transmission of the previous segment. The station error bit position, if set, would have denoted a lack of correct response by a served device when it was accessed by an "ACCD" or "TLKD" message during the previous segment. The least significant bits are remote interrupts assigned on a requisition basis.

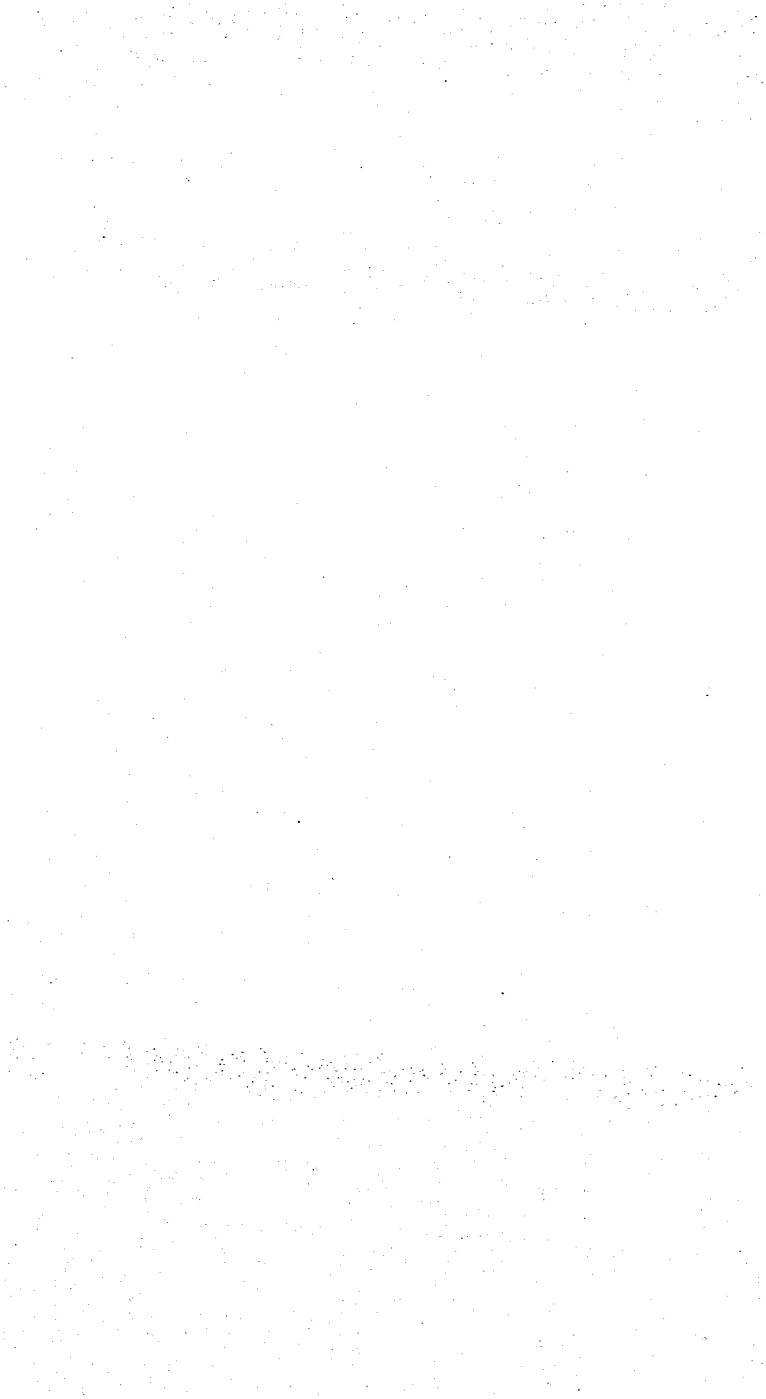
We have mentioned error reporting and error detection mechanisms throughout this section. The architecture of CSH is built upon a foundation of comprehensive error detection. Error detection is based on the 5-bit CRC field. It is transmitted inverted (incorrect) and is reinverted to be correct by the addressed slave. This constitutes a mechanism to detect a failed or an off-line slave. The CRC check also checks message integrity with the following error comprehension:

1) All single burst errors of 5 bits or less.

2) 95% of all single burst errors over 5 bits.

Additionally, there are message bit integrity checks performed at all repeaters that will flag illegal bit sequences. Finally, the master station compares a received "CMDBYT" against its transmitted "CMDBYT" counterpart. The net result is that it is virtually impossible to complete a faulty message transmission without detection of that faulty message.

Error detection is followed by message retransmission as a mechanism of error correction.



--SECTION 2--
DESCRIPTION AND OPERATION

MASTER STATION OPERATION

A block diagram of the master station appears in Figure 2-1. The physical layout and board content are shown in Figures 2-3 and 2-4 respectively. The cabling which ties the master station into the CSH system is shown in Figure 2-5.

The host interface stores status and data arriving to and from the PMC. It also buffers control lines controlled by a micro-program in the PMC that defines the nature of the data held. The nature of the stored data can be one of the following:

* RAM Memory address

* RAM Memory data

* Status

The direction of transfer is also indicated by the control lines.

The controller senses these control lines and directs the data between proper sources and destinations. Message format information for each message is deposited in the lower half of memory. The data destined for the INFO WORD of that same message is stored in a related location of the upper half of memory. Thus, the contents of memory messages are totally accumulated in RAM memory. The status lines tell the controller when a message burst or segment is ready for transmission.

The controller, then, senses the inputs from a message timing functional block. This block supplies timing signals which are synchronized with out-going messages. The controller synchronously accesses memory and deposits message formats into the Xmit register. The Xmit register serializes the information. The serialized information modulates the outgoing bit stream. The CRC generator concurrently computes the CRC check field. At the end of 24-bit period, the serialized CRC information modulates the bit stream. Then, for the next 22 bit times, the Fill generator modulates the bit stream.

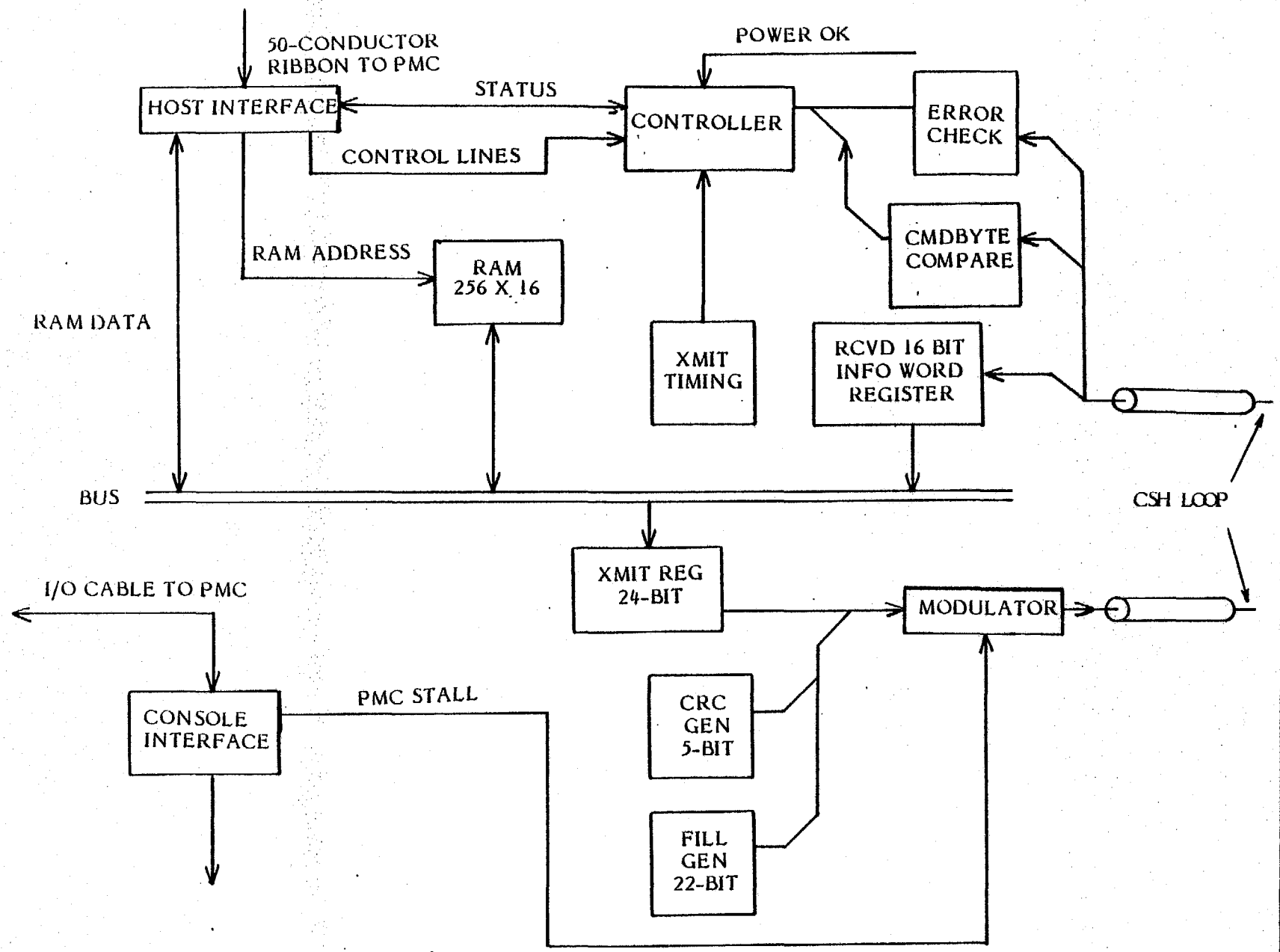
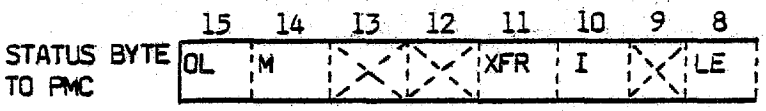


Figure 2-1. Block Diagram of DS3820 MMP Master Station.



OL - ON LINE (Hardware has relinquished message control to software)
M - Master mode request
XFR- Message burst transmission complete
I - Slave station demand encountered
X - Not used
LE- Transmission aborted because of link error



SS - SET SLAVE mode (Used by highway test module)
D - Set demand bit into idle message
GO - Transmit burst out of RAM memory
3 - Select loop 3
2 - Select loop 2
1 - Select loop 1
COM- Acknowledgement of the "M" bit above (PMC software has recognized master mode request)

Figure 2-2. Status Bytes Passed Between PMC and Master Station

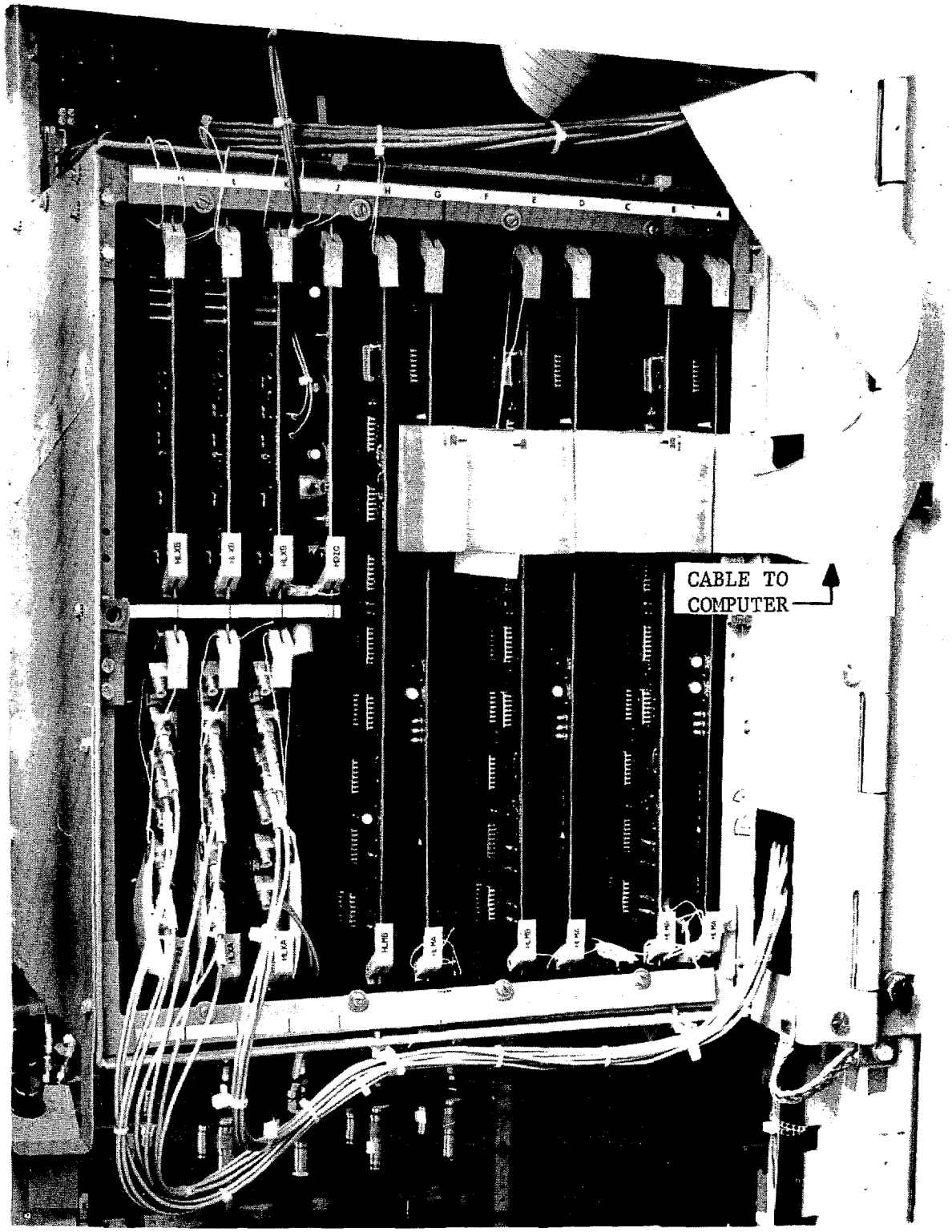


Figure 2-3. Master Station (3 Loops)

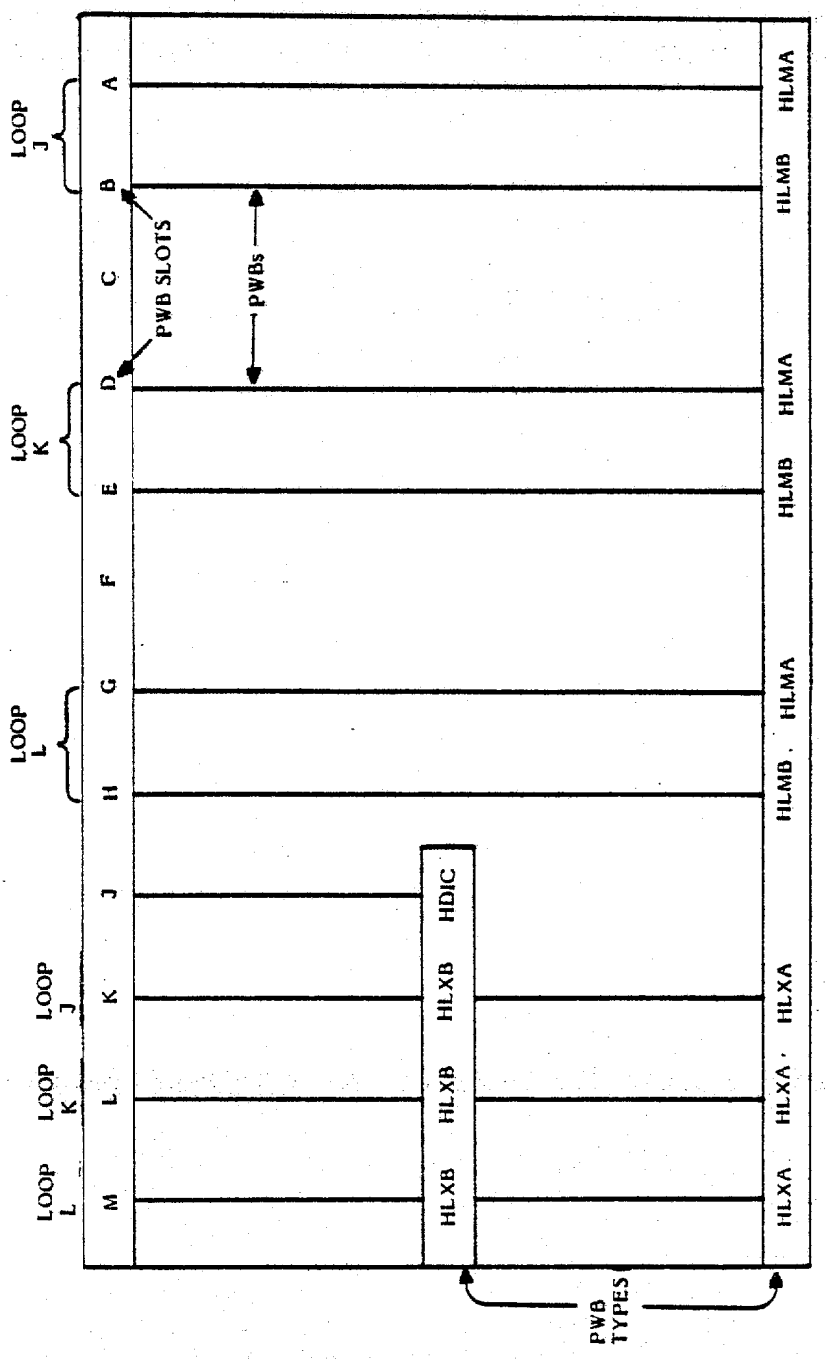


Figure 2-4. Master Station Layout

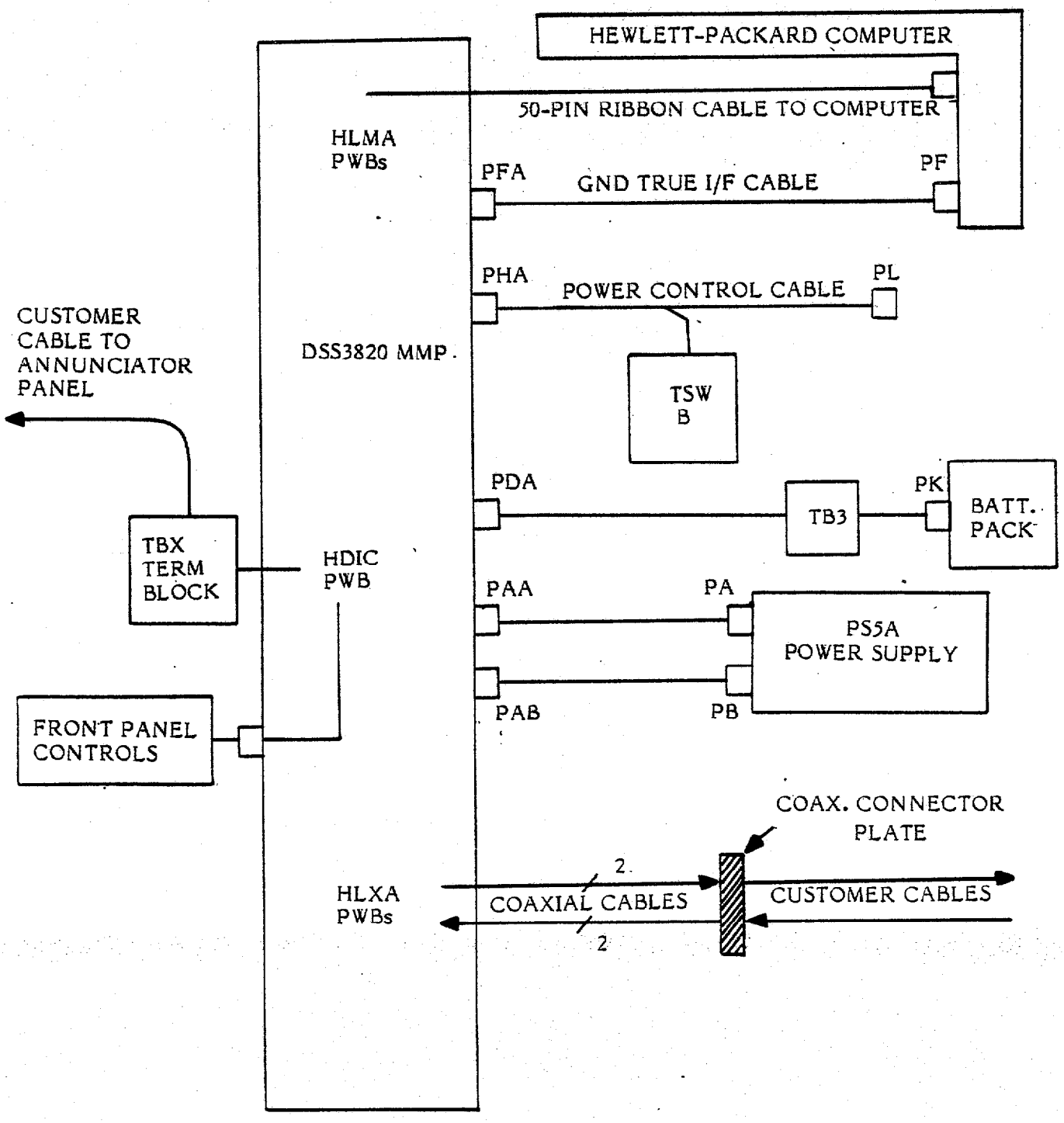


Figure 2-5.
Master Station Cabling Diagram.

At loop return, the Xmit timing block notifies the controller that enough time has elapsed for a message to arrive at loop return. The message, by this time, has been captured in the INFO WORD register block and in the CMDBYT compare block. The controller, then deposits the contents of the INFO WORD register into Random Access Memory (RAM). When reception of the entire segment is complete, the controller prepares a status bit indicating this, and interrupts the PMC. (See Figure 2-2 for the interpretation of status bytes passed between a PMC and master station.) The PMC then directs the transfer of the data portion of RAM into the PMC, thereby completing the acquisition of information from the CSH.

During the receipt of messages, the CMDBYTE Compare function checks the transmitted CMDBYTE against the received CMDBYTE. The CRC checker verifies the check result performed on the message stream. If an error is detected, a retry of the failed message is attempted. If the retry also fails, a link error is reported by status bit and the message transmission is aborted. Simultaneously, an interrupt is sent to the PMC.

The Console Interface board, HDIC, located between the CPU and the PMC console (see Figure 2-1) consists of the electronics which serves the switches and indicators of the PMC console pictured in Figure 1-5. Information enters the PMC via a Hewlett-Packard "Ground True" I/O board (Cat. # 12566B). There are also two fail-safe mechanisms implemented within the safety circuitry:

1. If the program of the PMC fails and thereby allows stall time-out, the modulation will be cut off on the CSH.

2. The power supply serving the DS3820 MMP has a pair of voltage-in-tolerance leads, one of which must be at logic 1 and the other at logic 0 (1LVSH and 0LVSL respectively). Should the power supply go out of tolerance, the ON-LINE status of the master station will not be honored and the software needed to create CSH message activity will be dormant.

SLAVE STATION OPERATION

The function of a slave station is to recognize a message addressed to it, and then to process the message according to its intent. A functional block diagram of a typical CSH slave station (DS3820 TM2) is shown in Figure 2-6. Such a slave station presents the controller with 4 bits of data defining the intent of the message passing through. These 4 bits are ignored unless the slave-addressed line becomes active. The controller then makes a decision as to which message-processing algorithm is to be invoked. During this message processing, the serial INFO WORD extracted from the message may be serial-to-parallel converted and then applied to address decoders or data drivers, depending upon the message being processed.

The addressing information contained in INFO WORD may activate the reduced bus address drivers, the full bus address drivers, or neither of them. This is controlled by address straps on the two HPBC boards which implement the reduced and full bus drivers respectively.

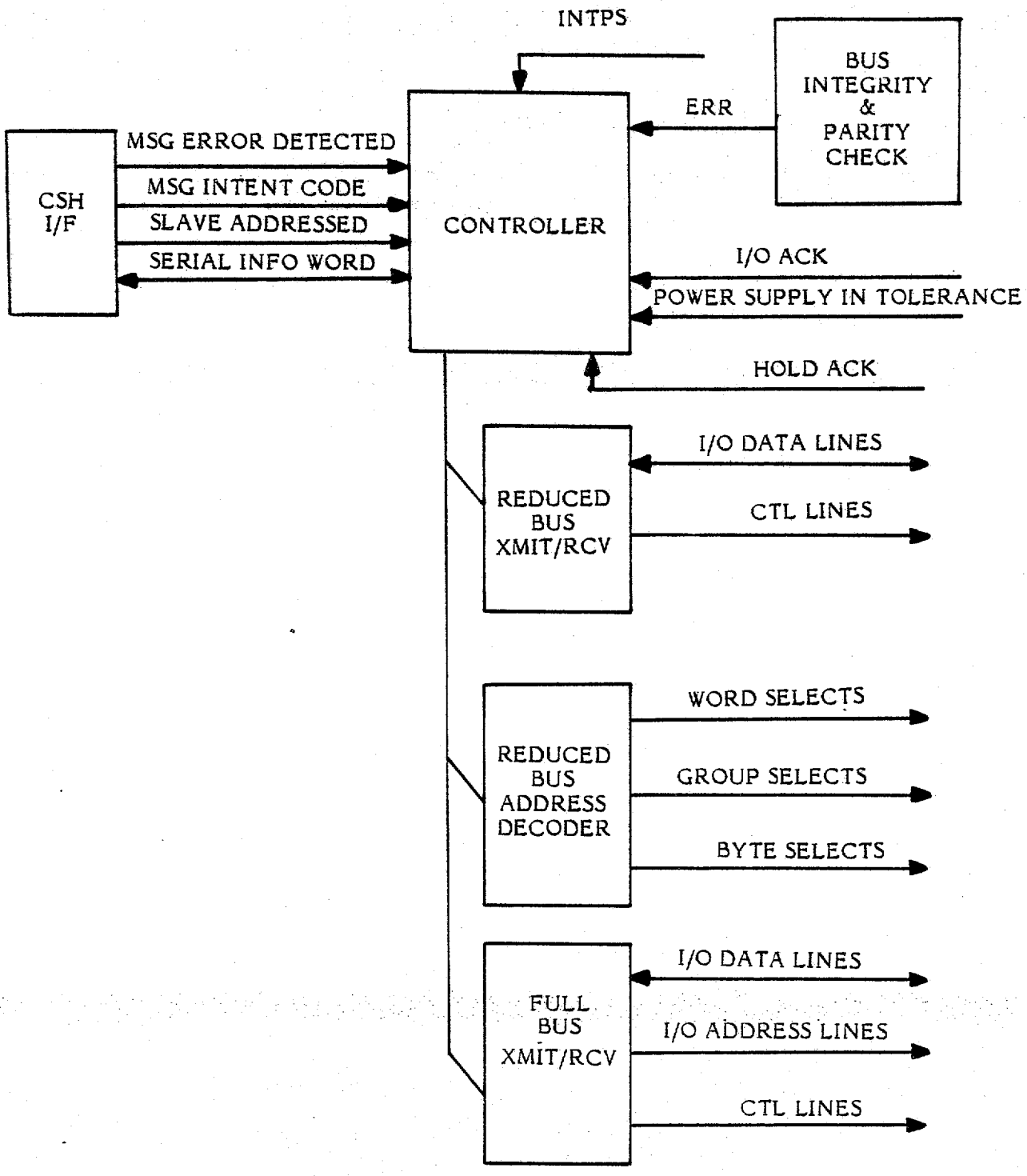


Figure 2-6.
Block Diagram DS3820 TM2 Slave Station.

If one of the HPBC boards becomes active, it relays a "HOLD ACK" signal to the controller. The controller then enables the data lines along with defining control lines. An I/O card recognizing the data lines being made available to it will respond with a "DATA ACK" signal. When the DATA ACK signal is recognized by the controller, the data transfer is completed. The data transfer is between a register residing within the controller and an I/O device attached to the I/O data lines.

The DS3820 MMI slave station functions similiarly to the DS3820 TM2 slave station described above. The difference is that I/O cards are self-contained within the DS3820 MMI itself.

Another difference is that there is a decoded address patchboard (HACA PWB) which routes group and linear select addresses to each I/O PWB. You will usually find the DS3820 MMI station applied in operators stations whereas the other type of slave station (DS3820 TM2) is usually connected to motor drives or switch gear equipment.

A functional block diagram of a DS3820 MMI slave station is shown in Figure 2-7 and a picture of such a slave station is shown in Figure 2-8. The physical layout is shown in Figure 2-9.

The controller shown in Figure 2-6 increments the I/O address register contained within it upon receipt of each "TLKD" or "ACCD" message addressed to the slave. This address register is also over-written by each "TLKA" or "ACCA" message recognized by the slave station.

The message error line is sensed by the controller. No output data is given the I/O if one of these errors occurs. Also, a link error status bit is set within the controller.

The bus integrity and parity check block shown in Figure 2-6 monitors data parity. If incorrect, a station error status bit will be set within the controller. The bus integrity circuits check for shorts on the I/O data bus and if one is found, they will keep the drivers inactive. The station error status bit in the controller is also set.

An input PWB in slot 1A of the DS3820 MMI slave station has 8 direct data outputs available as interrupts to the controller. The presence of one or more of these interrupts will cause the controller to assert its Demand bit within the "IDLE" message. The controller will insert the interrupt bits themselves as the low byte of the INFO WORD in an "INQ" message. Direct Data outputs are outputs that are exclusive to the I/O PWB and are not tied to a bus in parallel with other I/O PWBs.

There is a safety mechanism employed at all slave stations. It is invoked when modulation on the CSH loop is missing beyond 20 milliseconds. Such an event causes the slave station to either hold all I/O outputs at their last value or to reset them to zero. The choice is determined by a wire on the slave station back plane installed at the factory. It is normally wired to cause the reset of outputs.

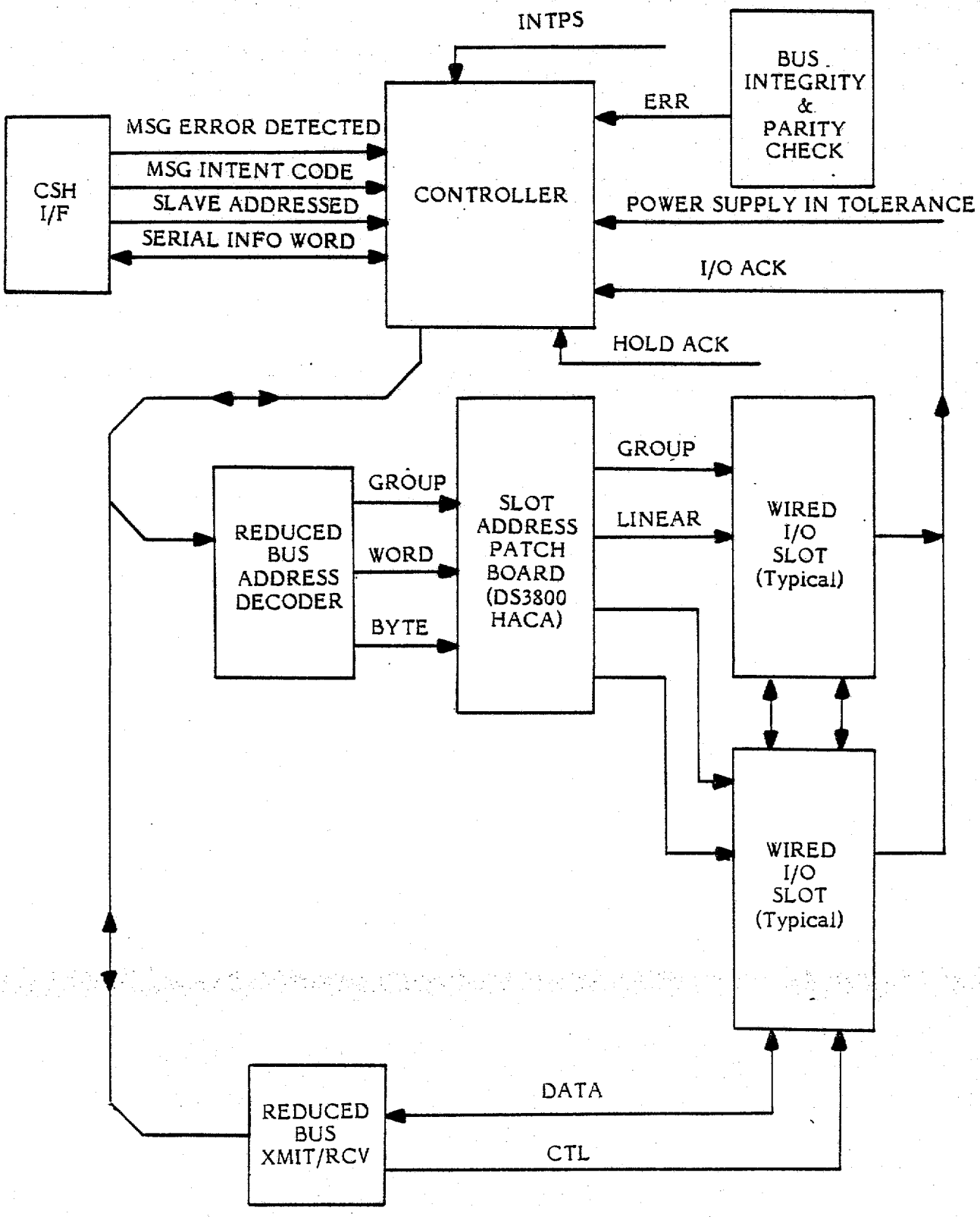


Figure 2-7.
Block Diagram DS3820 MMI Slave Station

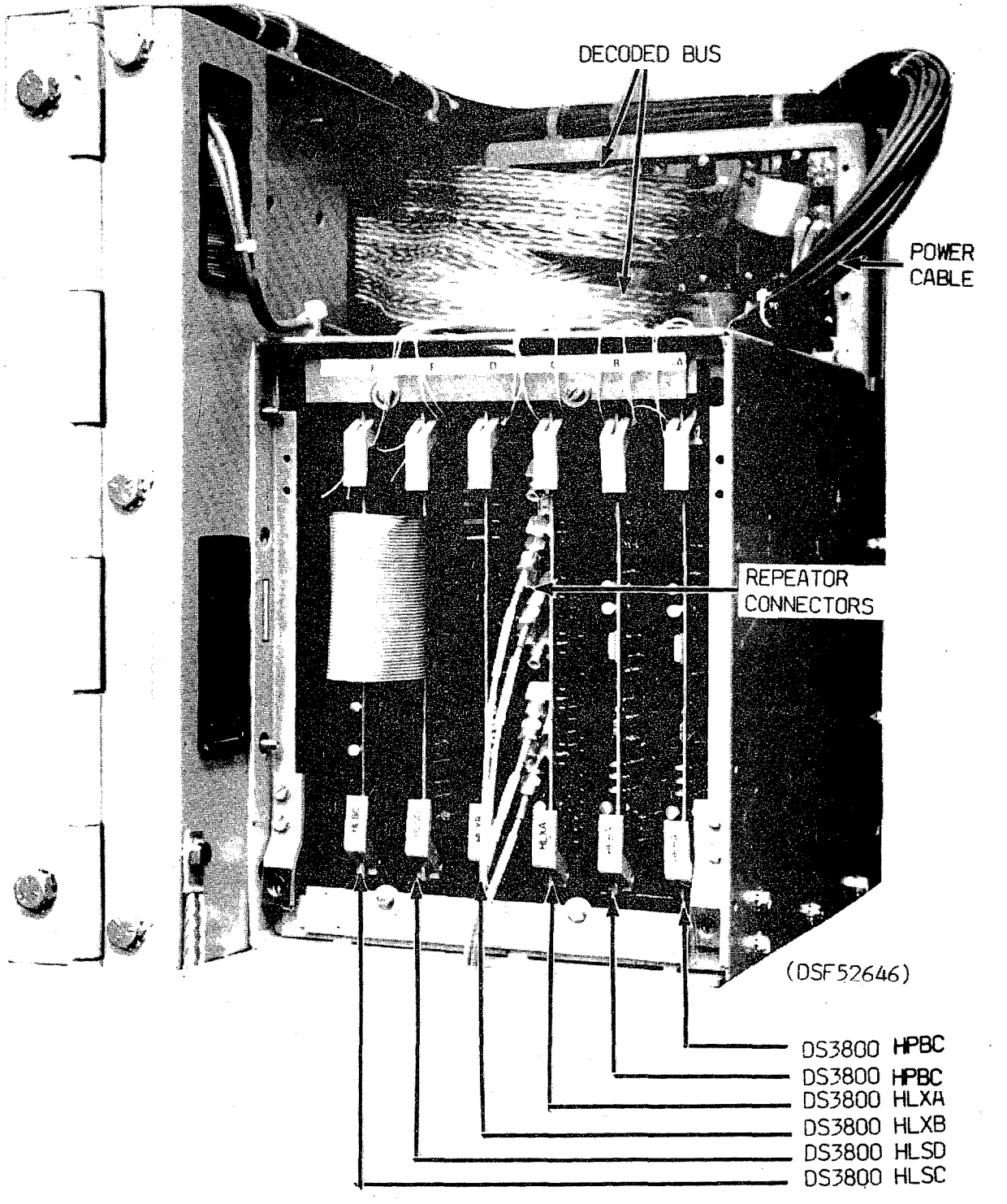
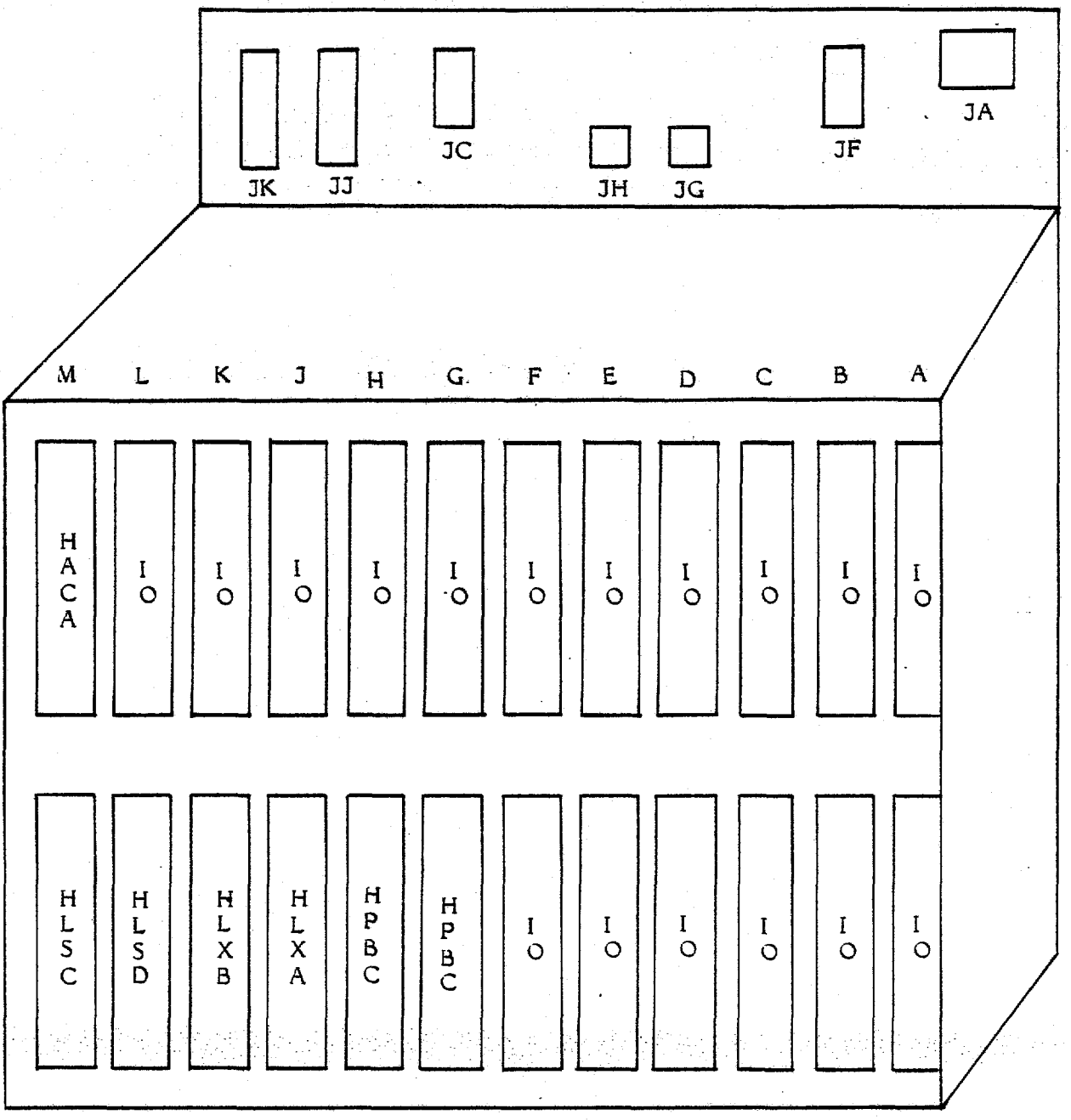


Figure 2-8.
DS3820 TM2 Slave Station



JJ & JF ORIGINATE DECODED I/O BUS
JK & JC ORIGINATE UNDECODED I/O BUS
JH,JG CABLE BETWEEN THEM BRINGS DIRECT DATA OUT OF I/O SLOT 1A TO CSH INTERRUPT INPUTS AT SLOT 2L
JA POWER SUPPLY CONNECTION

Figure 2-9.
DS3820 MMI Physical Layout

--SECTION 3--
INSTALLATION INSTRUCTIONS

****
* WARNING *
****

DO NOT PERFORM THE FOLLOWING PROCEDURE UNLESS ALL DRIVES ARE DISABLED FROM RESPONDING TO CSH-TRANSMITTED MESSAGES. OTHERWISE, PERSONNEL NEARBY THE PROCESS COULD BE INJURED AND EQUIPMENT COULD BE DAMAGED.

INITIAL INSTALLATION

The following procedures should be performed during the initial installation phase of any CSH loop to insure the integrity of the system.

1. Terminate the customer cables by installing connectors per assembly instructions given in Figure 3-1.

2. When installing CSH cabling in open cable trays or conduit, avoid routing coaxial cables through a cable tray also containing AC power wiring.

3. Connect customer cables to form the required CSH loops according to the configuration shown in Figure 3-2. If it becomes necessary, for any reason, to lengthen the cable, wrap cable-splicing connectors with electrical tape to prevent possible ground loops.

4. Assume that installation variables, i.e., jumpers on various PWBs, are installed correctly. However, you can become familiar with these variables by referring to Figures 3-6 through 3-13 and the accompanying tables.

5. Slave stations are normally installed with I/O connections intact. Assume that the I/O is already intact. Be sure to ground cabinets to earth ground at the cabinet ground lug.

6. Turn on power at each slave station. Also, switch each repeater into the CSH loop via SW1, located on the repeater (DS3800 HLXA).

7. Verify that the 1LVSH and OLVSL signals at JA-1 and JA-2 of the slave power supplies are at logic 1 and logic 0 respectively. If not, the supply voltages are out of tolerance and will require adjustment.

8. Set the Highway Test Module (HTM) to I/O SIMULATION mode and connect between LOOP START and LOOP RETURN as shown in Figure 3-3. Turn on the Highway Test Module and set it for REPEAT mode of operation. The HTM RECEIVED SIGNAL indicator must flash repetitively. If it does not do so, the CSH loop is open-circuited.

9. If the CSH cable is open-circuited, walk the loop in the direction of LOOP RETURN, beginning at the LOOP START. As long as the cable is whole, the LKACT light-emitting diode (LED) on the HLSC board will be lit. (Refer to Figure 3-10 and Table 3-5 for location and significance of LKACT LED.)

10. Repeat procedures 8 and 9 (above) as applicable to all primary and redundant loops.

11. Verify that the cabling at the master station is connected as shown in Figure 2-5, and that the card arrangement corresponds to that of Figure 2-4.

12. Short TP1 to TP2 on the DS3800 HDIC board.

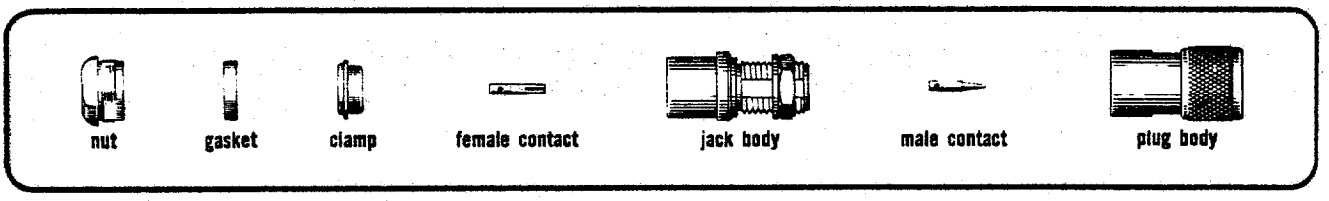
13. Disconnect the Highway Test Module from the CSH loop, then reconnect loop input and loop return to the master station per Figure 3-3. Connect the redundant loop input and loop return to the master station per Figure 3-3.

14. Turn on the master station AC line power and the master station power supply. Verify that the MST LED on the HLMA board is lit (See Figure 3-12). If it is not lit, you are attempting to operate the loop from the wrong PMC. (The loop appears as a slave to the wrong PMC).

15. Connect an oscilloscope to TP2 of the HLMA board. This test point will show either the waveform displayed in the photograph of Figure 3-4 or the waveform shown in Figure 3-5. The former indicates the presence of idle messages, i.e., the loop has been properly synchronized. The latter waveform, if present, indicates that loop synchronization is being continually attempted and cannot be achieved. This indicates the existence of a fault somewhere along the loop.

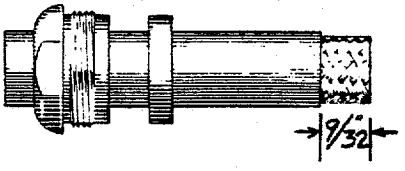
16. Repeat steps 12 through 15 for each loop master station. This completes the checkout of all primary and redundant CSH loops. Leave the test set-up as it is for the post-installation checkout.

An exploded view of the detailed parts of the connector to be assembled is as follows:

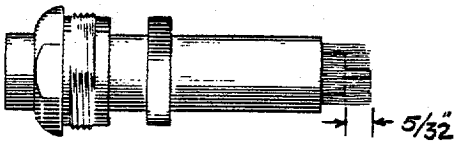


ASSEMBLY INSTRUCTIONS:

1. Place nut and gasket over the cable with the "V" groove toward the clamp, and cut off 9/32 inch of the jacket as follows:



2. Comb out braid and fold out. Cut off cable dielectric to 5/32 inch as shown below:



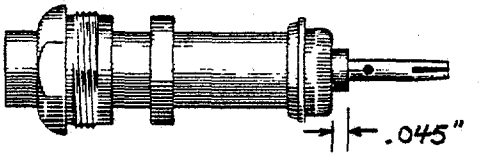
3. Pull braid wires forward and taper toward center conductor. Place clamp over braid and push back against cable jacket as follows:



Figure 3-1. Attaching Connector to Customer Cable.

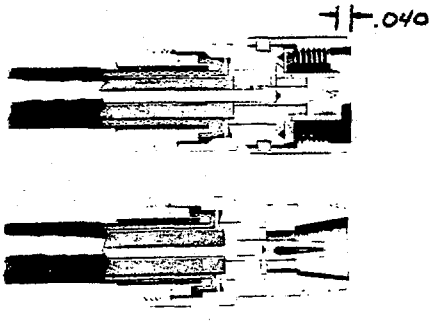
Figure 3 (Continued)

4. Insert cable and parts into connector body. Ensure that the sharp edge of clamp seats properly in gasket. Tighten nut.



NOTE

The dimension of spacing between the center pin and end-of-connector is essential to the proper functioning of the connector. Therefore, the tolerance shown is critical.



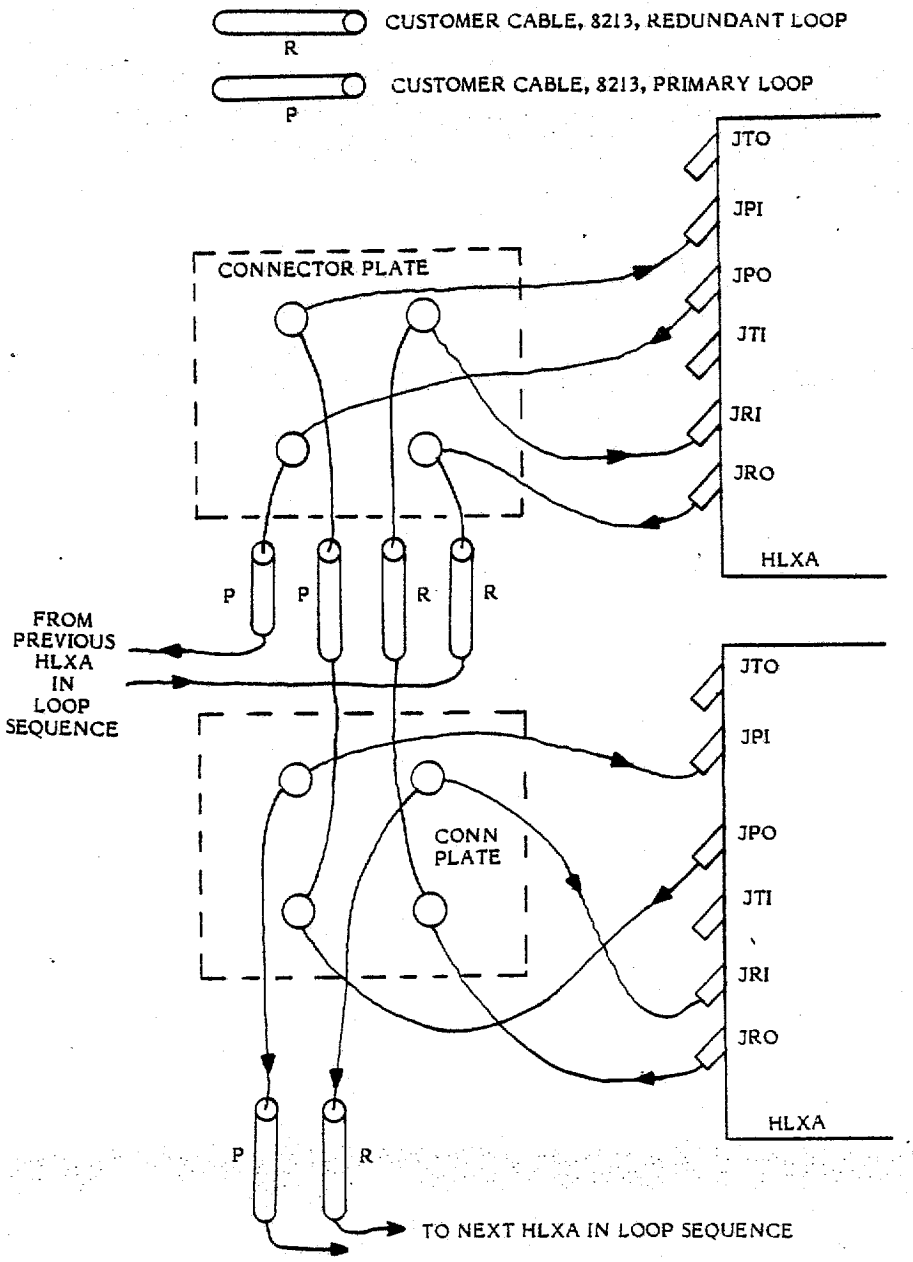
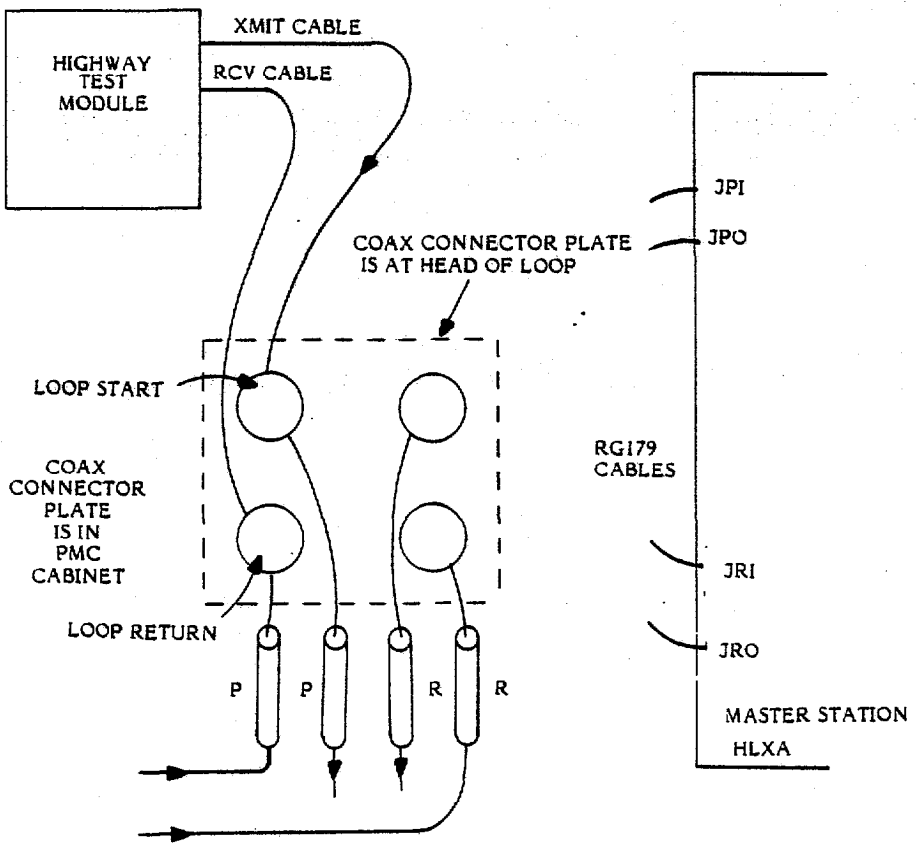


Figure 3-2.
Coaxial Cable Installation Diagram



DISCONNECT RG179 CABLES AT PLATE AND THEN CONNECT HIGHWAY TEST MODULE IN THEIR PLACE TO PERFORM TEST. REPLACE RG179 CABLES AFTER TEST.

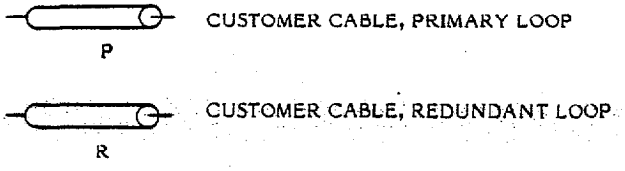


Figure 3-3.
Test Set-up Loop Continuity

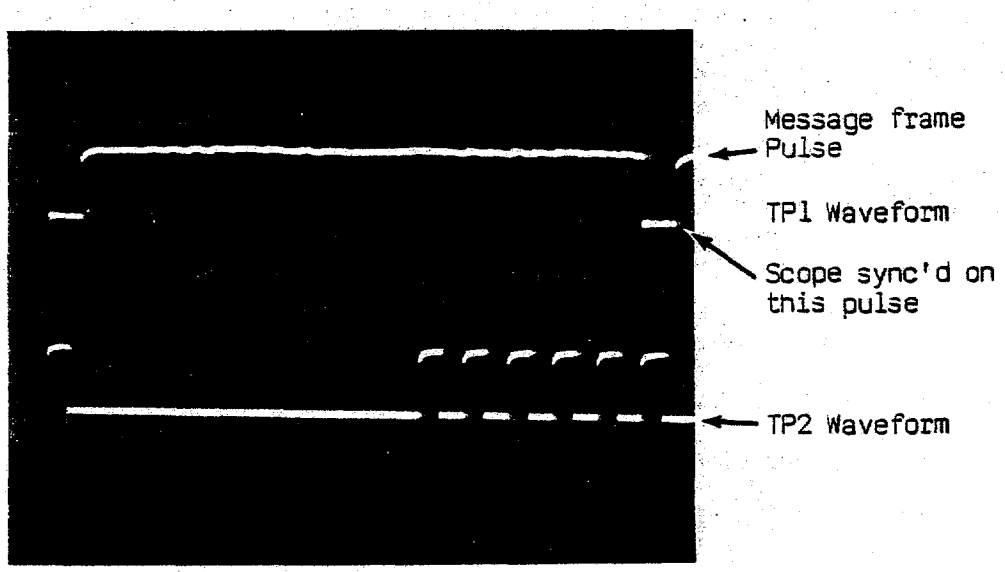


Figure 3-4. Appearance of Idle Waveform

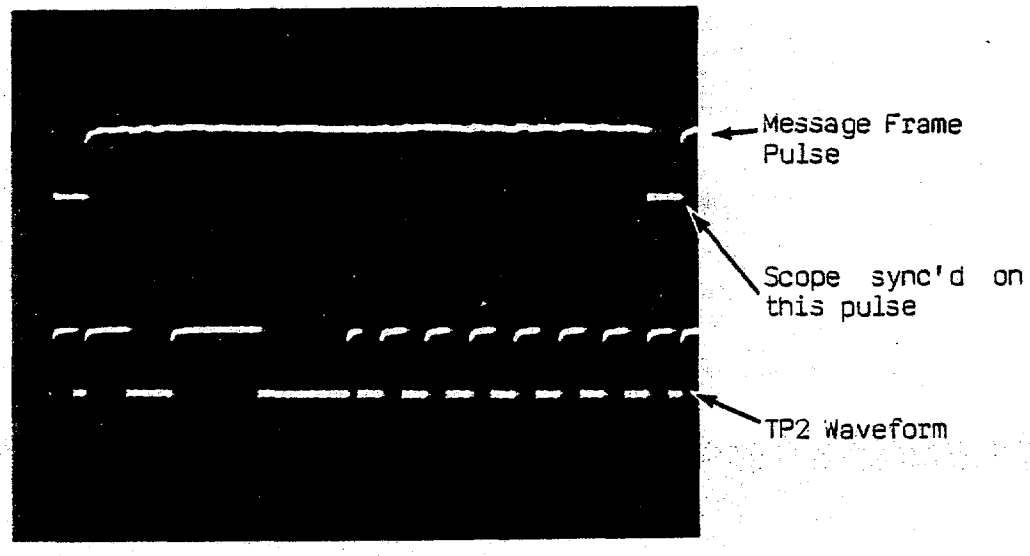


Figure 3-5
Appearance of Sync Waveform

POST INSTALLATION CHECKOUT PROCEDURE

The post-installation procedure verifies the capability to automatically switch over from primary to redundant cable. The procedure will also verify that the entire CSH system, including the DS3840 PMC and connected slave stations, are operational.

1. Disconnect the primary cable connected to LOOP RETURN on the coax. connector plate (See Figure 3-3).

2. Verify that the PCBL indicator on the HLMA board (Figure 3-12) goes out. Also verify that all LKACT indicators on HLSC boards (Figure 3-10) remain steadily lit.

3. Disconnect the redundant cable at LOOP RETURN on the coax. connector plate.

4. Verify that the PCBL indicator on the HLMA board is presently blinking at a random rate as DS3820 MMP tries to establish viability of one cable or the other but finds neither cable to be acceptable.

5. Reconnect the primary cable at LOOP RETURN on the coax. connector plate.

6. Verify that the PCBL indicator is currently glowing steadily, then reconnect the redundant cable at LOOP RETURN on the coax. connector plate.

7. Repeat steps 1 through 6 (above) for each CSH loop in the system.

8. Down-load the system operating tapes into the central processing unit (CPU).

NOTE

The CIAD program is an interpretive language program for directing CSH message activity. The CIAD program is included on the tapes mentioned in step 8 (above).

9. Load a CSH checkout tape (including CIAD program) which has been prepared to exercise all slave stations and connected I/O PWBs served by the slaves.

10. Place the CSH system off-line at the PMC control panel, then execute the CSH checkout program. If a malfunction occurs, print-outs on the CRT screen will indicate the addresses of involved equipment and the type of error detected will appear in English text.

11. Remove jumper at TP1 and TP2 of HDIC printed wiring board upon successful completion of item 10, above.

Table 3-1.
HLXA Indicators and Installation Variables

PWB: DS3800 HLXA

TYPICAL APPLICATION: DS3820 TM2 of Figure 2-8, DS3820 MMI of Figure 2-9 and DS3820 MMP of Figure 2-4.

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
Jumper	As shown in Fig. 3-6.	Slave will automatically switch to redundant cable when required. The slave will not switch when the jumper is in the NO position.
JTO, JTI	--	These are coax. connectors for Highway Test Module.
JPO, JPI	--	These are coax. connectors for primary cable connection.
JRO, JRI	--	These are coax. connectors for redundant cable connection.
SW1	UP Position	Repeater is switched into CSH loop.
	DOWN Position	Coax. bypasses repeater. Also connects repeater to connectors JTI and JTO.
"RUN" LED	--	This is the indicator showing that the repeater is switched into the CSH loop.

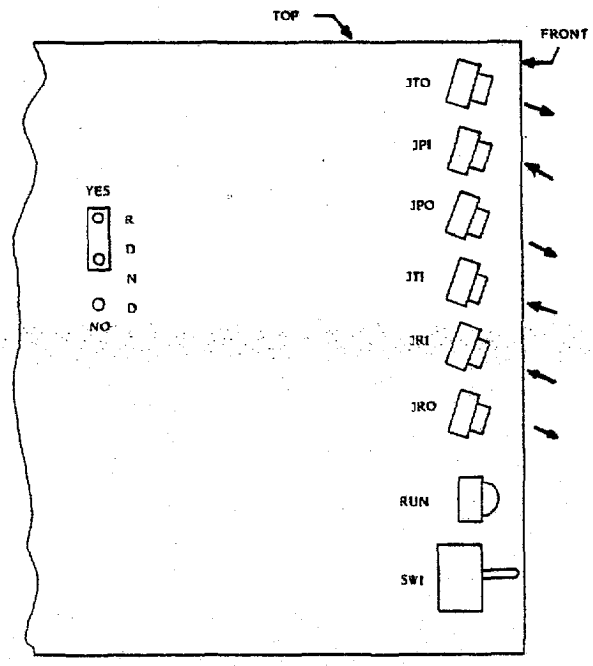


Figure 3-6.
DS3800 HLXA Installation Features

Table 3-2.
HLXB Indicators and Installation Variables

PWB: DS3800 HLXB of Figure 3-2

TYPICAL APPLICATION: DS3820 TM2 of Figure 2-8,
DS3820 MMI of Figure 2-9, and
DS3820 MMP of Figure 2-4.

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
Jumpers 1,2,4,8	As shown in Figure 3-7	Station jumpered to slave address "A" hex. Jumpers have no meaning in a DS3820 MMP operating as a CSH master station. When the DS3820 MMP is operated as a slave, the number imprinted next to the jumper's active position corresponds to the binary encoding of the station's slave address.

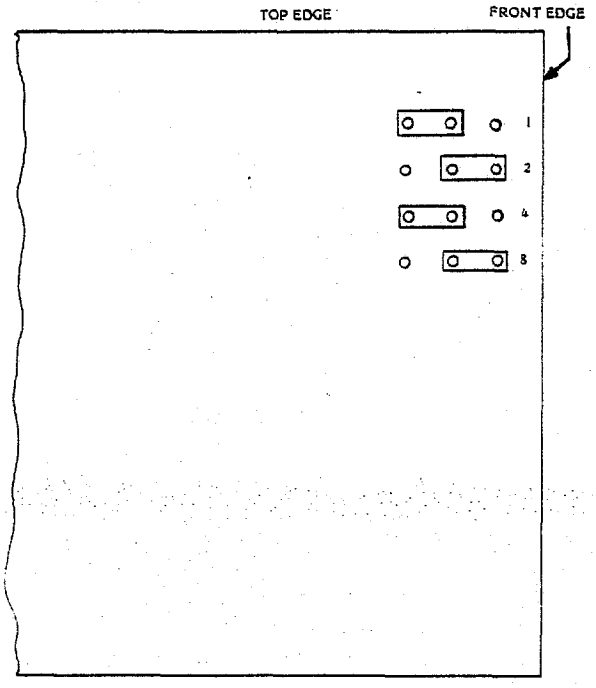


Figure 3-7.
HLXB installation Variables

Taole 3-3.
HDIC Indicators and Installation Variables

PWB: DS3820 HDIC of Figure 3-3.

TYPICAL APPLICATION: DS3820 MMP of Figure 2-4

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
CR3 LED	ON (Amber)	Temperature within PMC cabinet is within limits.
CR2 LED	ON (Amber)	The PMC program is not stalled
CR1 LED	ON (Amber)	Battery voltage is within limits
TP1, TP2	--	Connection of a test lead between these test points is required to defeat a stall. There is no modulation on the CSH during a stall.
		CAUTION DO NOT DEFEAT STALL DURING NORMAL CSH OPERATION, OTHERWISE, CERTAIN SAFETY FEATURES WILL BE DEFEATED, i.e., MODULATION CUTOFF DURING STALLOUT, OR WHEN THE MASTER STATION POWER SUPPLY FAILS.
SW1	MOMENTARY	Lamp test switch for CSH control panel, CR1, CR2 and CR3.
SW2	MOMENTARY	Annunciator contacts held closed.
	UP	Annunciator contacts held open.
CONN	--	Connection point for cable between remote annunciator and activating contacts residing on this PWB.

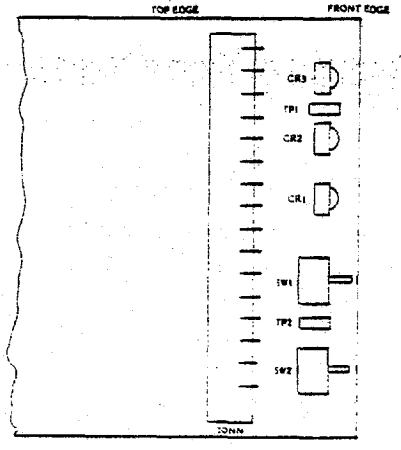


Figure 3-8.
DS3800 HDIC Installation Features

Table 3-4.
HPBC Indicators and Installation Variables

PWB: DS3800 HPBC of Figure 3-4.

TYPICAL APPLICATION: DS3820 TM2 of Figure 2-8, and DS3820 MMP of Figure 2-4.

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
Range Jumpers	As shown on Figure 3-9.	HPBC responds only to I/O addresses encoded as XX = 11 (See Figure 3-4) used to select between multiple I/O buses served by one slave station.
"RST" LED	ON	I/O cards served by HPBC are being held reset
"BUS OK" LED	ON (Amber)	I/O bus not shorted.
"POK" LED	ON (Amber)	Initially off. The LED latches on with the first good I/O transfer and will always reflect the status of the latest I/O transfer handled.
BUF DRV Jumper	As shown on Figure 3-9	The HPBC card is in the decode mode, i.e., it heads reduced bus to I/O.
AUTO Jumper	--	Causes HOLD ACK to be automatically returned as handshake. (See Section 2) It is commonly used in conjunction with a reduced bus.
SW1	UP	Momentary I/O reset applied
SW1	DOWN	Forced I/O reset applied
SW2	MIDDLE	No I/O reset applied

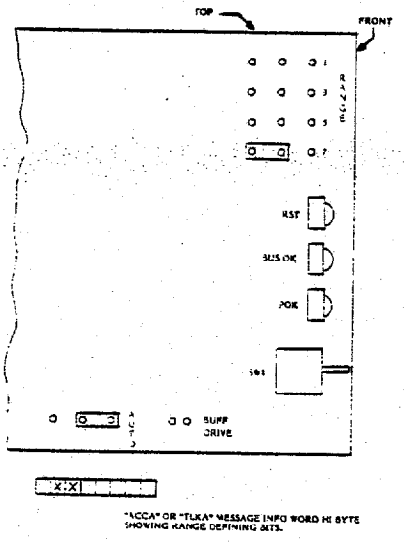


Figure 3-9.
DS3800 HPBC Installation Features

Table 3-5.
HLSC Indicators and Installation Variables

PWB: DS3800 HLSC of Figure 3-5

TYPICAL APPLICATION: DS3820 TM2 of Figure 2-8, and DS3820 MMI of Figure 2-9

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
"LOK" LED	ON (Amber)	Comes on during power-up. Turns off when a faulty message is detected. It can only be reset by an "INQ" message once it has come on.
"STAOK" LED	ON (Amber)	Comes on during power-up. It turns It can only be reset by an "INQ" message once it has come on.
LKACT LED	ON	Legitimate modulation is passing through the CSH station.
STACT LED	ON	Messages addressed to this station are being received.

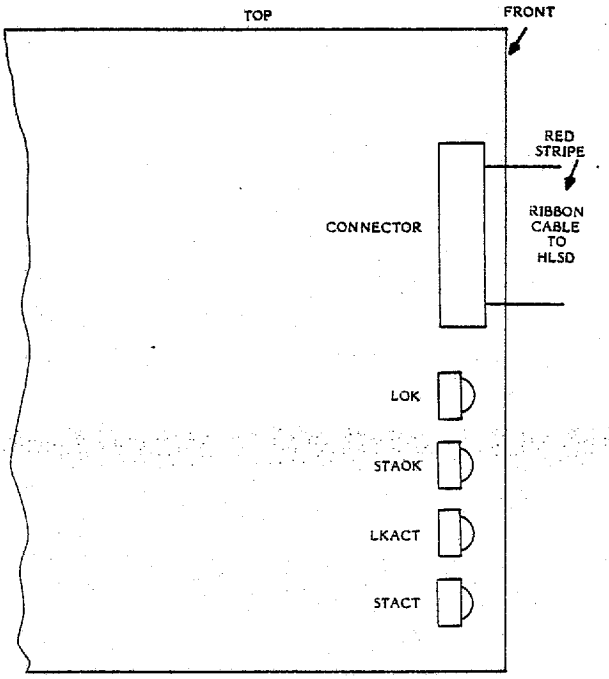


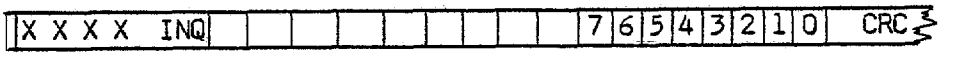
Figure 3-10.
DS3800 HLSC Installation Features

Table 3-6.
HLSD Indicators and Installation Variables

PWB: DS3800 HLSD of Figure 3-6.

TYPICAL APPLICATION: DS3820 TM2 of Figure 2-8, and DS3820 MMI of Figure 2-9

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
Jumper J0	Shown on Figure 3-11	INTPT 0 (Bit 0) is selected for positive edge triggering.
Jumper J1	Shown on Figure 3-11	INTPT 1 (Bit 1) is selected for negative edge triggering.
Jumper J2	Shown on Figure 3-11	INTPT 2 (Bit 2) is selected for positive edge triggering.
Jumper J3	Shown on Figure 3-11	INTPT 3 (Bit 3) is selected for negative edge triggering.
Jumper J4	Shown on Figure 3-11	INTPT 4 (Bit 4) is selected for positive edge triggering.
Jumper J5	Shown on Figure 3-11	INTPT 5, 6, 7 (Bits 5, 6, and 7) are selected for positive edge triggering.



CSH MSG

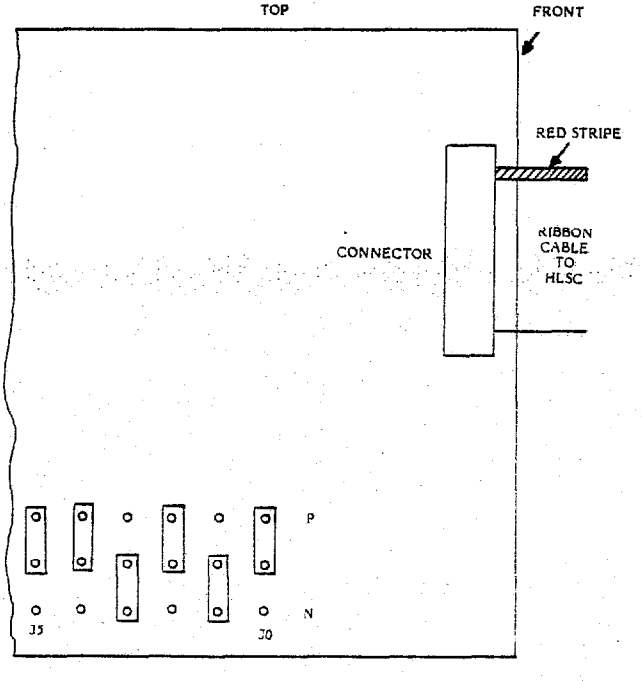


Figure 3-11.
DS3800 HLSD Installation Features

Table 3-7.
HLMA Indicators and Installation Variables

PWB: DS3800 HLMA of Figure 3-8

TYPICAL APPLICATION: DS3820 MMP of Figure 2-4

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
Jumper L5	Shown on Figure 3-12	Jumper must be as shown when PWB is installed in slot. "A" of Figure 2-4 must be at other position at all other slots.
Jumper L4	Shown on Figure 3-12	A CSH station, when in the role of a master, will not allow switching between primary and redundant cables. When in the opposite state, the switching is allowed to take place. The jumper has no effect when operating in slave mode.
Jumper L3	Opposite of that shown in Figure 3-12	Loop III is enabled for servicing by PMC
OR		
Jumper L2	Opposite of that shown in Figure 3-12	Loop II is enabled for servicing by PMC
OR		
Jumper L1	As shown on Figure 3-12	Loop I is enabled for servicing by PMC
MST LED	ON	Station hardware has assumed the role of master. The PMC software's ability to support mastership is not indicated.
PCBL LED	ON	Primary cable in service. This has no meaning if station is functioning as a slave.

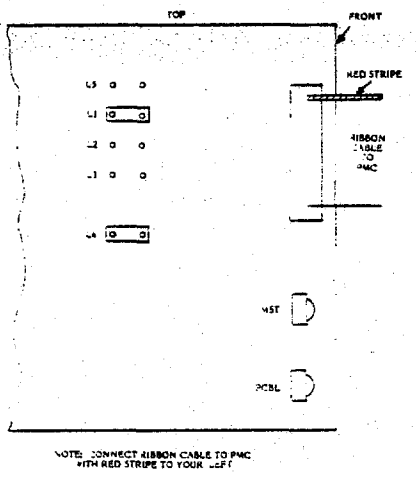


Figure 3-12.
DS3800 HLMA Installation Features

Table 3-8. HLMB
Indicators and Installation Variables

PWB: DS3800 HLMB of Figure 3-8

TYPICAL APPLICATION: DS3820 MMP of Figure 2-4

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
STA ACTIVE LED	ON	Station operating as a slave, is receiving messages addressed to it. This has no meaning if acting as a master station.
SLAVE JUMPER	AS SHOWN ON Figure 3-13	Station configured to function as a CSH slave.

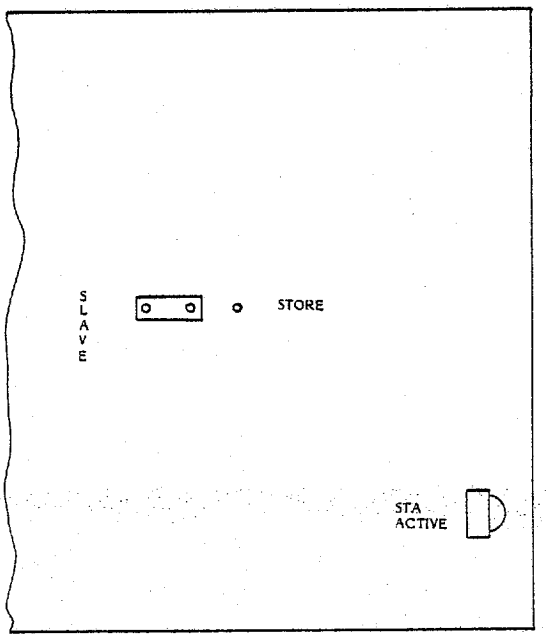


Figure 3-13.
DS3800 HLMB Installation Features.

--SECTION 4--
CSH OPERATING INSTRUCTIONS

CONTROLLING THE CSH

****
* *
* CAUTION *
* *
****

This equipment is designed to operate in a normal industrial atmosphere. Operating radio transmitters nearby may cause the control equipment to misoperate. It is recommended that radio transmitters not be operated within the vicinity of the control equipment when the panel doors are open, nor at any time within the confines of any power control room.

CSH message segment transmission is controlled by software within the PMC. The interface to that software is at the PMC control panel. (See Figure 4-1).

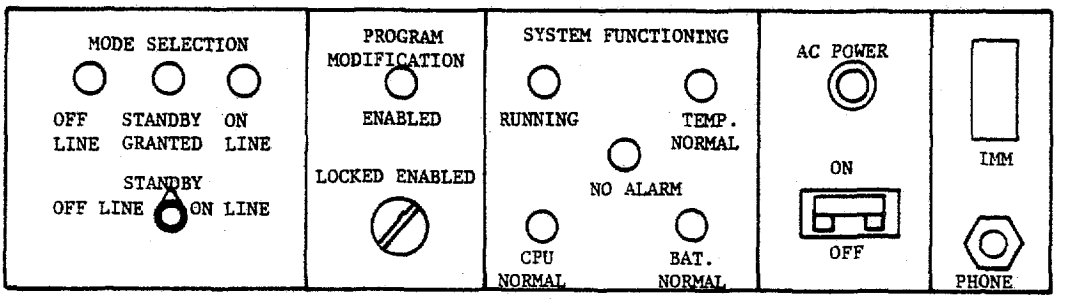


Figure 4-1.
PMC Control Panel Layout

NOTE

CSH segments are sets of messages which run in a specific time slot for the purpose of acquiring or disseminating control variables and/or I/O information.

The MODE SELECTION switch has three positions: OFFLINE, STANDBY, AND ON LINE. When set to OFF LINE, there is no modulation on the CSH unless test points TP1 and TP2 on the HDIC PWB are jumpered together. In off line mode, messages are sent out on the CSH which merely reads inputs from the drive(s).

To load CSH system software, set the MODE SELECTION switch to OFF LINE position and set the PROGRAM MODIFICATION Key switch to the ENABLED position. After system software has been loaded, set the MODE SELECTION switch to ON LINE. The CSH should automatically start operating. This results in the running of process-initializing CSH segments followed by process-controlling CSH segments. If no segments appear to be running (no STA ACTV indicators lit at the slave stations) it may be because there were no CSH segments initialized for running when the software was first loaded. However, if the CSH segments are active, it is possible that the software will turn off the CSH, i.e., if the uncorrectable error rate exceeds 2 errors per second. This could result from the appearanace of errors originating in a slave station(s). Such errors could be a result of the incorrect set-up of the slave station or the drive/I/O which it serves, i.e., wrong address strapping, wrong cabling, wrong PWB installed slave station power supply out of tolerence, or master station power supply out of tolerance.

****
* *
* WARNING *
* *
****

THE MODE SELECTION SWITCH SHOULD NOT BE SET TO ON LINE POSITION UNLESS THE SYSTEM DRIVES AND DEVICES HAVE FIRST BEEN PLACED IN A FAIL-SAFE CONDITION. THIS ACTION IS NECESSARY BECAUSE WHEN THE SYSTEM IS ON LINE, OTHERWISE THE OUTPUT OF CONTROL VARIABLES WILL BE ACCEPTED AND ACTED UPON (POSSIBLY HARMFULLY) BY THE DRIVE(S).

There is a PROGRAM MODIFICATION Key switch with LOCKED and ENABLED positions. When the key switch is set to the ENABLED position, the information in the program can be modified from a nearoy CRT terminal acting through PMC software. Additionally, when the PROGRAM MODIFICATION switch is set to the ENABLED position, a test program (CIAD) can be run for the purpose of testing the CSH when faults are suspected. While in the ENABLED Key switch position, the red ENABLED light will be on. The MODE SELECTION switch must also be in the ON LINE position in order to allow CIAD interpretive programs to exercise the CSH as described in Section 5.

System Functioning

There are five amber lights which allow the operator to monitor various functions of CSH operation:

NO ALARM Light - This is a combination light and switch. If the light goes off, it signifies that an alarm exists. That alarm can be acknowledged by depressing the cap of the NO ALARM light. If this light goes out while the system software is functioning, indicator lights on the front panel of the PMC computer will display the condition that caused the NO ALARM light to go out. This can be relied upon only during system operation.

RUNNING Light - When on, this light indicates that the CSH is running and not stalled out. The CSH can stall out if the PMC software does not reset the stall timer before it times out. The time-out period ranges from 6 to 378 milliseconds and is software selectable.

TEMP. NORMAL Light - If this light goes out, it indicates that the PMC cabinet is too hot and the system should be safely shut down.

BAT. NORMAL Light - Not applicable.

CPU NORMAL Light - When on, this light signifies that the independent power supplies of the PMC computer are within acceptable limits.

--SECTION 5--
MAINTENANCE AND TROUBLESHOOTING

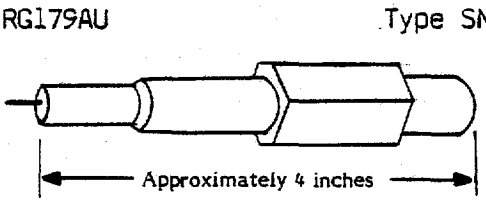
RECOMMENDED TEST EQUIPMENT

* Tektronix, Model 485 oscilloscope or equivalent.

  Requirements are: Bandwidth 50 MHZ (Minimum)
  Delayed sweep capability
  Dual channel display
  Chopped sweep capability
  External triggering capability

* Coaxial Cable Stub (Useful for observing modulation at JTO or JTI on the HLXA board.)

  Cut from spare cable to form stub as illustrated below:



* Highway Test Module

* Digital Logic Analyzer with 20 nanosecond resolution (Minimum)

* Digital Voltmeter
  Useful for measuring power supply voltages. Also can be used for measuring ohmic continuity of coaxial cables.

PREVENTIVE MAINTENANCE

We recommend that the following preventive maintenance procedures be performed at least once each 30 days:

1. Verify that the PCBL indicator (located on the HLMA board) at the head of each CSH loop, is lit. If this indicator is not lit, there has probably been an electrical break in the primary cable. Refer to the procedure for restoring primary cable operation in "Troubleshooting Symptom #1", below.

2. Verify that all threaded coaxial cable connections are at least finger tight.

TROUBLESHOOTING PROCEDURES

Any one of the following symptoms warrant that follow-on troubleshooting procedures be implemented:

1. PCBL indicator off or intermittent when the HLMA PWB (in DS3820 MMP) is operating in master mode.

2. Failure of CSH to come up on line or remain on line.

3. Excessive error rate notification at the IMM screen.

4. NO ALARM indicator goes out at less than half hour intervals.

Troubleshooting Symptom #1 (above).

Symptom. PCBL indicator off or intermittent when the HLMA PWB (in DS3820 MMP) is operating in master mode.

Probable Cause. There has recently been a break in the electrical continuity of the primary cable.

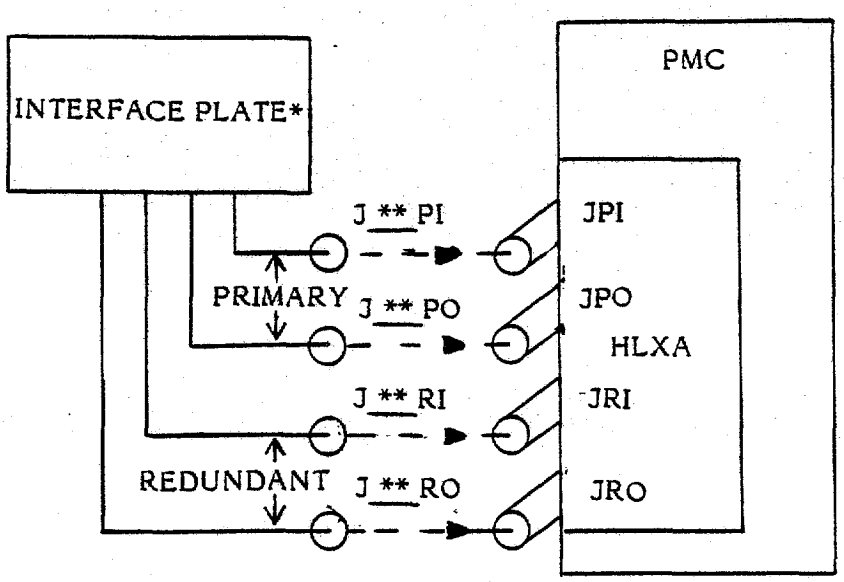
Recommended Fault Finding Procedure. Proceed as follows:

1. Disconnect all drives and devices from CSH.

2. Disconnect primary and redundant cables at the head of the CSH loop as illustrated in Figure 5-1.

3. Observe that the PCBL indicator now flashes erratically because the master station should be alternately trying to establish viability of one or the other of the cables. If no flashing is observed, replace the HLMA or HLMB card.

4. Reconnect the primary cable. If the PCBL indicator is lit (not intermittent), the primary cable is continuous (not open-circuited). Regardless of the condition of the primary cable, proceed to step 5.



*The interface plate contains the coaxial connectors which interface customer cables with PMC or slave station equipment.

NOTE: Disconnect interface coax cables from JPO & JPI and from JRO & JRI as shown.

**The second letter designation of the connector will be A, B or C, depending on the requisition.

Figure 5-1
Test Connections for Automatic Cable Switch-over

5. Walk the loop in the direction of the loop return (See Figure 3-3). Check the coaxial cable connections at each connector plate for tightness of connection. Examine each HLXA board to verify that its redundant (RDND) jumper is in the YES position. Check the LKACT indicator of the HSLC board at each slave station. If it is not lit or is intermittent, it indicates the existence of a cable break between that station and the preceding station.

6. After the primary cable is verified to be operational, you should confirm the viability of the automatic cable switchover function as follows:

Using an ohmmeter, check the redundant cable for continuity.

Reconnect the redundant cable.

Disconnect the primary cable momentarily. The PCBL indicator must extinguish and remain continuously unlit. Reconnect the primary cable.

Momentarily disconnect the redundant cable. The PCBL indicator should come on and remain on. Reconnect the redundant cable.

After all foregoing checks have been satisfactorily completed, return the CSH system and all associated drives to on-line status.

Troubleshooting Symptom #2

Possible Causes. The CSH has failed to come up on line or to remain on line for one of the following reasons:

Errant software

Defective PMC

Defective HDIC or Hewlett Packard 12566B boards

Defective master station boards HLMA or HLMB

Abnormally high rate of interrupts to the PMC caused by unsatisfiable slave demands

Poorly seated or incorrectly oriented ribbon cable between HLMA and PMC

Recommended Fault-Finding and Corrective Procedure. Proceed as follows:

1. Disconnect drives and devices from CSH.

2. Connect jumper between TP1 and TP2 of the HDIC board to defeat the modulation cut-off feature of the CSH.

3. Using a logic analyzer, check for the presence of the signals listed in Table 5-1. If any of the signals are missing, replace the suspected faulty components (as recommended in Figure 5-1) on a one-at-a-time trial basis. Attempt on-line operation after each trial replacement.

Table 5-1.
Principal PMC/CSH Handshaking Signals

SIGNAL NAME	SIGNAL TYPE	PWB/PIN	MEANING OF SIGNAL	PROBABLE FAULTY COMPONENTS IF SIGNAL MISSING
1WST	Pos.Pulse	HLMA/17	Status Byte is arriving from PMC	PMC, Ribbon cable, HLMA, 12566, HDIC or CSH driver program is not functioning.
1CKST	Pos.Pulse	HLMA/20	Contents of status byte FROM PMC is about to be checked.	HLMA, HLMB
1INTHP	Pos.Pulse	HLMB/27	Alerts CSH driver program during initialize, End of transmission, error discovery, demand discovery, Can overwhelm software if continuous.	HLMA, HLMB, if missing. Slave if continuous
OACT	Neg.Pulse	HLMA/23	PMC needs immediate attention from DS3820 MMP serving this loop for information purposes.	HLMA, PMC
OSWAK	Neg.Pulse	HDIC/26	PMC console switch change acknowledge.	
1LKRST	Neg.Level	HLMB/80	Permits only sync and idle messages.	Power supplies
OSTRST	Pos.Level	HLMA/32	Permits modulation	Software not functioning.
OGO	Neg.Pulse	HLMA/16	Permission to transmit message segment	HLMA, PMC, HDIC, Software
OXFR	Neg.Pulse	HLMA/8	Transmission of message Segment complete	HLMA
1RST	Pos.Pulse	HLMA/18	Requesting status from DS3820 MMP first expected Handshake after each 1INTHP	HLMA, HLMB, PMC, Ribbon cable, HDIC, 12566B, software

4. Continuous occurrences of interrupts (several appearances of 1INTHP) can result in clogged software and could prevent the CSH from coming up on line. Should this be suspect, turn off the slaves one at a time using SW1 on the HLXA card at each slave. If the continuous interrupts disappear, a defective slave has been located. Troubleshoot the slave using the Highway Test Module.

5. Should all of these procedures be exhausted without success, it may be necessary to call your GE Service Representative for assistance.

6. When CSH operation becomes successful, you should run a CIAD test program to check the operation of each loop. Section 8 explains the use of the CIAD program.

7. Remove the HDIC jumpers and restore the drive for responsiveness to CSH signals.

8. Turn the system on-line at the CSH console.

Troubleshooting Symptom #3

Symptom. There is an excessive error rate as reported oy message on the IMM screen.

Recommended Troubleshooting Procedure. Proceed as follows:

1. Disconnect drives and devices from responding to CSH-supplied signals.

2. Write and execute a CIAD-interpretive program that exercises the suspect functional area. Examples of CIAD-interpretive programs are located in Section 8. However, if the fault is so severe as to incapacitate the CIAD program, proceed to step 3. It may also be necessary to check the key handshake signals of Table 5-1.

3. An alternate method of troubleshooting is to use the Highway Test Module instead of the CIAD program, or in conjunction with the CIAD program to locate the faulty component. The Highway Test Module operates in any of the following modes which are discussed below:

I/O Mode/Monitor Function
I/O Mode/Simulate Function
Slave Mode
Master Mode

I/O Mode/Monitor Function--merely displays messages destined for the address dialed by the Highway Test Module operator's panel. This has limited diagnostic potential.

I/O Mode/Simulate Function--can extensively test a slave station of the DS3820 TM2 or DS3820 MMI variety. It should be used when the IMM CRT display indicates faults stemming from a slave station implemented by one of the DS3820 modules.

Slave Mode--can extensively test a DS3820 MMP module operating in slave mode. It should be used when the IMM CRT display identifies a slave address implemented by DS3820 MMP hardware as being suspect.

Master Mode--is used to exercise a PMC in its role of hosting a CSH loop. The master mode is used in conjunction with the MTEST commands of the CIAD interpretive language. It cannot function alone. The PMC supplies different message sequences dependent upon the MTEST command executed. The Highway Test Module acts as an intelligent slave operating in an unconventional manner relative to CSH protocol. In this role, the Highway Test Module strips certain information from some test messages and returns this stripped information plus fault analysis information in other test messages. The PMC will analyze the returned information and then display diagnostic information on the IMM screen. See the MTEST explanations of Section 8 for more detail. The Highway Test Module operates as a slave station, address "F" hex code when in master mode. The connection diagram for wiring the Highway Test Module into CSH loops appears in GEK-65151.

Corrective Action For Symptom #3. The usual cause of a high alarm rate is the corrupting of information being passed between the PMC and the RAM located on the HLMA board. This may be corrected by replacing the HLMA board and/or the HLMB board to decrease the alarm rate. If that fails, replace the PMC. A less likely possibility is that the HDIC board or the 12566 interface board is faulty and needs to be replaced.

Troubleshooting Symptom #4

Symptom. NO ALARM indicator goes out at less than half hour intervals.

Recommended Troubleshooting Procedure. Observe the Hewlett Packard (H.P.) computer console. Interpret the H.P. display according to the following information.

PMC Alarm Indications--

Alarm conditions detected by the PMC are indicated by several means. An indication that an alarm condition exists is given by turning off the NO ALARM indicator on the PMC console. A detailed identification of the alarm is displayed on the 16 display indicators on the PMC computer. These indicators are numbered 0-15, right to left. Table 5-2 lists the individual alarm conditions associated with each indicator and the appropriate operator actions.

Alarm indicators are cleared by depressing the NO ALARM pushbutton on the PMC console. Clearing alarms in this manner also resets the "down" condition of certain external devices, causing them to re-try previously failed operations.

Certain serious faults, i.e., a burst error rate in excess of 2 per second, cause the PMC to go off-line. This fault can only be corrected by performing a restart sequence, that is, by turning the PMC MODE SELECTION "REQUESTED" switch to the OFF LINE position, then back to the ON LINE position. In extreme cases, initializing may not work. In this case, you must reload and/or troubleshoot.

The left-hand half of the display (indicators 8-15) display CSH error condition details. Those conditions which are serious enough to warrant taking the PMC off line will also cause indicators 0, 1, and 2 to come on.

Table 5-2.
PMC Alarm Indications

PMC INDICATOR ON	MEANING	OPERATOR ACTION
0	A restart may be necessary to clear the alarm condition	Restart the PMC by turning the PMC MODE SELECTION "REQUESTED" switch to the OFF LINE position and then to the ON LINE or STANDBY position.
1	A reload may be necessary to clear the alarm condition.	Reload the PMC. If both restart and reload are indicated, the restart should be attempted first.
2	CSH data link fault. The PMC has been taken off line due to a serious CSH data link fault.	Note the error. Restart or reload as specified by indicators 0 and 1.
3	FSS data link fault.	Clear alarms then retry current operation.
4	IMM fault. This is most probably a mini-cartridge tape problem.	Clear alarms, then retry current tape operation.
5	Printer fault. The PMC printer (if present) is not in a ready condition.	Correct printer problem. Then clear alarms to resume operation.
6	Operator station fault. The PMC operator station is not in a ready condition.	Correct operator station problem. Then clear alarms to resume operation.
7	Spare	
8-14	CSH fault condition. May or may not be serious enough to degrade process.	If accompanied by indicator 2, take action as for indicators 0 and 1. If not accompanied by indicator 2, see information below to interpret meaning (X means indicator on).

INDICATOR(S) 14	13	MEANING	OPERATOR ACTION
0	X	Channel J fault	Interpret indicator 12
X	0	Channel K fault	Interpret indicator 12
X	X	Channel L fault	Interpret indicator 12

12					
0				Station Error*	Interpret indicators 11-8 as station number.
X				Channel	Interpret indicators 11-8 as error type below.
10	11	9	8		
0	0	0	0	CSH Master fault	Correct problem, then take action per indicator 2.
0	0	0	X	Link error**	If not accompanied by indicator 2, this is a spurious error. Frequent occurrences indicate pending serious problems. Perform maintenance on master/coax. cables.

*NOTE: A station error occurs when a threshold of 0 errors or more per second occur. If the error rate decreases to 5 errors per second or less, a station error is no longer indicated.

**NOTE: If 2 or more errors per second occur, the system goes off line.

11	10	9	8		
0	0	X	0	Excess link error rate.	Serious version of previous error. Will also be accompanied by indicator 2. Attempt restart and if this error is repeated, use the HTM to isolate the problem.
0	0	X	X	Channel off-line	Spurious error, sometimes accompanying link error. See link error operator action.
0	X	0	0	Illegal interrupt	Can occur spuriously during power fail/auto-restart. Usually not serious. If it persists, replace master hardware and restart.
0	X	0	X	Illegal control word	Reload and restart.
0	X	X	X	Master communication error	Restart. If problem reoccurs, replace master hardware.

Repair Procedures

The only CSH repair procedure consists of either replacing malfunctioning boards or open-circuited sections of coaxial cables around the CSH loop.

Spare Parts

A Renewal Parts quotation book containing parts and material lists and catalog numbers is supplied with the Control Signal Highway. Some complete elements and assemblies may be returned to the General Electric Renewal Parts Service Center for repair and return.

A Directo-Matic^{R} element exchange service is available for DS3800-type printed circuit cards (those used in the CSH).

All parts and repair orders should be placed with your local G.E. sales representative.

--SECTION 6--
CIAD PROGRAM

CIAD is a high-level language interpreter which enables the user to exercise and troubleshoot the Control Signal Highway. The high-level language is used to compose message segments placed onto the CSH by the PMC. CIAD, then interprets and executes that language in real time.

There are two trouble-shooting features that are important and only available in conjunction with the CIAD program:

1. The status passed from CSH to PMC can be read by CIAD (through its SG variable) and placed on the IMM CRT display,and

2. The PMC can be exercised to ensure that it is capable of hosting CSH (through its MTEST commands). This requires also the use of the Highway Test Module.

NOTE A

You should be familiar with the Hewlett Packard operating system RTE-M described in Hewlett Packard publication 92604-90002. Otherwise, you may not understand the syntax in the following designated instructions.

NOTE B

In the following instructions, CIAD will respond with a "#" prompt, ready to accept interactive commands where indicated.

CIAD LOADING INSTRUCTIONS

CIAD may be included as a part of the utility package which is loaded using the FSS "FLOAD" function. See FSS Operator Manual (GEK-65435) for details. Alternately, CIAD may be loaded from cassette tape into a free partition of suitable size. Instructions for cassette loading are as follows:

a. Ensure that a partition of suitable size is available (must be 9 pages or more). If necessary, a function which is not needed may be removed to make room with the following command:

OF,XXXXX,8 (CR) See Notes A,B (above)

b. Insert the cartridge tape containing CIAD into the cassette drive.

c. Use the APLDR function to load CIAD.

*LO,LU,,,P# (CR) See Notes A,B

d.When CIAD has been loaded, type:

*RU,CIAD (CR)

CIAD will respond with a "#" prompt ready to accept interactive commands.

COMMAND SYNTAX

You communicate with CIAD using commands or procedures which have the following general form:

#COMMAND, p1, p2, p3, ...,pn (CR)
|
└──── CIAD interactive prompt

Each command is designed to perform a specific function. The meaning of each parameter following the command name is dependent upon the command used. Whether parameters are source or destination parameters is determined by the command they are used with and their position in the parameter string. Parameters may be given in one of five forms:

a. A signed or unsigned DECIMAL CONSTANT in the range -32768 to 32767. Decimal constants are only useful as source parameters.

b. A signed or unsigned OCTAL CONSTANT in the range -100000B to 077777B. Octal constants are denoted in the parameter list by entering a "B" suffix after each constant. Octal constants are only useful as source parameters.

c. An unsigned HEXADECIMAL constant in the range 0H to FFFFH. Hex constants are denoted in the parameter list by entering an "H" suffix after each constant. Hex constants are only useful as source parameters.

d. A symbolic INTERNAL variable which is an internal program variable. If its context implies that it is a source parameter, its current value is used as the parameter. If its context implies that it is a destination parameter, its current value is updated with the new data.

e. A symbolic EXTERNAL variable is a reference to one of the I/O devices (IMM or tape drives). If the context of the parameter implies that it is a source parameter, the data to be used as the parameter is obtained from the specified I/O device. If the context implies a destination parameter, the data is written to the specified I/O device.

INTERNAL VARIABLES

0G thru 9G - General purpose variables whose meaning is defined by the user.

J,K, and L - Variables containing the logical unit numbers representing the CSH loops.

IG - Variable containing the logical unit number of the IMM terminal.

LG,RG - Variable containing a code number determining the format of data entered or output via the EXTERNAL VARIABLE.

FG - The format codes are:

0 - Octal
1 - Decimal
2 - Hex
3 - ASCII

SG - Variable containing status about the last request results. Its format is shown below:

BIT	
15	1 = Master station on line
14	1 = Active master, 0 = passive master
13	
12	
11	1 = Transfer complete (should always = 1)
10	1 = Poll demand (should always = 0)
9	1 = Slave station error occurred as result of previous operation.
8	1 = Link error occurred as result of previous operation
7	
6	
5	
4	
3	
2	
1	
0	

EXTERNAL VARIABLES

External variables are used for transferring parameter information into (source) or out of (destination) the computer during the interpretation of the command line. The value of the FG variable specifies type of conversion to be used in converting from / to ASCII.

IMM	When used as a source parameter, this variable causes a " " prompt character to be output to the CRT. The user then enters numeric or ASCII data to be used as the parameter to the command. When used as a destination parameter, this variable outputs the data to the terminal screen.
LTAPE,RTAPE	When used as a source parameter, this variable reads the next record from the left or right cassette drive in ASCII and then converts it according to the current value of FG. When used as a destination parameter, the output data is converted to ASCII according to FG and written to the left or right cassette drive as the next record.

INTERACTIVE COMMANDS

The following symbolic conventions are used in the discussion of the CIAD commands:

H - denotes a hexadecimal command

iG - denotes an internal variable

xG - denotes an external variable

N - denotes a decimal constant

O - denotes an octal constant

pn - denotes the nth parameters in the parameter string

[,pn]- denotes an optional parameter

(CR) - denotes the carriage return key

All omitted parameters are equivalent to entering a zero constant for source and are not defined for destination parameters.

NOTE

In the following narrative, interactive commands are enclosed in a block.

A parameter is omitted by entering two successive commas with no non-blank characters between.

SET,p1 [,p_2] (CR)

Data from the source parameter p2 is transferred to the destination parameter. Legal parameters are O, N, iG, H, and xG.

ABORT (CR)

CIAD is turned off. It may be turned on again with *RU,CIAD.

ECHO [,p_1] (CR)

Commands are "echoed" (written) to the device whose logical unit number is specified by the source parameter p_1 as they are executed. Legal parameters are O, N, H, ig, and xG. The interpreter will stay in echo mode until it executes an ECHO, command. Legal parameters are IG, LG, RG or any constant provided it is a legal LU# for the system.

** (ASCII text up to 38 characters) (CR)

This command is used as a comment or prompt command and causes the command line itself to be written to the IMM screen.

CALCULATE, p_1,p_2,p_3 (CR)

This command is useful for performing arithmetic manipulations with globals or constant. p_2 is a special "operator" parameter entered as:

,+	-integer addition modulo 32768
,-,	-integer subtraction modulo 32768
,*,	-integer multiplication modulo 32768
,/,	-integer division modulo 32768 (result not rounded)
,AND,	-16-bit logical "AND"
,OR,	-16-bit logical "OR"
,XOR,	-16-bit logical "EXCLUSIVE OR"

When executed, the value specified by p_1 is replaced by the value obtained by performing the operation specified by p_2 on p_1 and p_3.

Legal parameters for p_1 and p_3 are O, N, H, iG, and xG.

i.e., CALCULATE ,OG,-,3 calculates the equation OG = Minus 3

RLINK,p_1,p_2 [,p_3 [,p_4]] (CR)

The previous command reads a 16-bit word from a station on the CSH loop specified by p_1. p_2 specifies the station address to be read. If the local PMC is a master of the p_1 loop, legal values of p_2 are 1 to 15 (0 to FH), provided these stations exist. In this case, the RLINK command sends a message sequence consisting of a "talk status" message which reads the slave status and resets any errors caused by the "talk address". The point address to be read is specified by p_3 and is any valid global whose value is from 0-255. The 16-bit data word returned from the slave station is stored at optional p_4. If p_4 is not specified, the returned variable is not acquired by the PMC.

If the local PMC is a slave to the p_1 loop, the value of p_2 should be zero. In this case, no message is sent; the local slave memory is read at point address p_3 and the data is returned to optional p_4. If p_4 is not specified, the returned information is not acquired by the PMC.

Legal parameters are 0, N, H, iG, and xG, subject to range checks mentioned above. It is the user's responsibility to verify that the specified station exists on the addressed loop.

WLINK,p_1,p_2 [,p_4]] (CR)

The command shown above writes a 16-bit word to a station on the CSH loop specified by p_1. p_2 specifies the station address to be written to. If the local PMC is a master of the p_1 loop, legal values of p_2 are 1-15 (0 to FH), provided these stations exist. In this case, the WLINK command sends a message sequence consisting of an "accept address" message followed by a "talk status" message which reads the slave status and resets any errors caused by the "accept address". The point address to be written is specified by p_3 and is any valid variable in the range 0-255. The optional variable P_4, supplies the data to be written to the point. The data are zeros unless p_4 is specified.

If the local PMC is a slave to the p1 loop, the value of p_2 should oe zero. In this case, no message is sent; the local slave memory is written from p_4 at the point address specified by p_3.

Legal parameters are 0, N, H, iG, and xG, subject to range checks mentioned above. It is the user's responsibility to verify that the specified station exists on the addressed loop.

TLINK, p_1, p_2 , [p_3 [,p_4]] (CR)

The command shown above writes a 16-bit word to the test register of a station on the CSH loop specified by p_1. p_2 specifies the station address. If the local PMC is a master of the p_1 loop, legal values of p_2 are 1 - 15 (0 - FH) (provided these stations exist). This command is not defined for p_1 designating a slave mode. An "accept test data" message is sent which writes the contents of p_3 to the test register. This is immediately followed by a "talk test data" message which reads back the test register contents into p_4.
Legal parameters are O, N, H, iG, and xG, to range restrictions mentioned above. It is the user's responsibility to verify that the specified station exists on the addressed loop.

ERATE, p_1 [p_2] (CR)

This command collects a line of error history information for the CSH loop specified by p_1 and outputs it to the device specified by p_2. If p_2 is IMM or omitted, the error information is displayed on the IMM screen. If p_2 is LTAPE or RTAPE, the error information is available for a master station:

1. Link Error Rate

   - The filtered number of link errors per second occurring on the channel.

2. Station Error Rate

   - The filtered number of station errors per second occurring on the channel.

3. - A list of stations which are currently "unhealthy". A station becomes unhealthy when it notifies the master of a station error while the station error rate exceeds the defined error rate limit for the system. Unhealthy stations become healthy again when the station error rate again falls to 1/2 the error rate limit.

When a channel is specified in p_1 that is a slave station at this PMC, only the CSH Error Rate is available.

Below is an illustration of a typical display line indicating a low CSH error rate, a high station rate, and several unhealthy slave stations.

#ERATE, J, IMM (CR)

.00010 75.300 1 2 5 A F

This function is meaningful only on PMCs having continuous segments. Errors caused by CIAD's messages are reported internally to the SG variable and do not contribute to the error rate.

MTEST, p_1,p_2 [,p_3 [,p_4]] (CR)

The above command is used to invoke one of seven pre-defined tests to be used in conjunction with the HTM (Highway Test Module). Refer to the HTM manual for details of the test procedures. The following discussion briefly outlines the purpose and form of each command.

p_1 -Specifies the channel to be tested (J,K, or L)
p_2 -Specifies the test to be performed (1 to 7)
p_3 -Is an optional mode parameter. Omitting it will cause one cycle of the test to execute. Entering "C" will select continuous cycle mode and will repeatedly execute the selected test until an error occurs.
p_4 -Is an optional error display parameter. Omitting it will cause a continuous mode test sequence to halt if an error occurs. Entering "F" will suppress errors. This is useful for scope troubleshooting of the master cards.

The explanations of the MTEST commands that follow contain references to Status Error, Verify Error and Interrupt Error.

Status Error means that one or more bits in the Status Byte acquired by the PMC from the HTM are incorrect.

Verify Error means that message contents acquired by the HTM do not agree with that sent by the PMC.

Interrupt Error means that the PMC did not receive a transmission-complete interrupt from the CSH.

MTEST COMMANDS

Test 1 - Slave mode memory test

Command: #MTEST, J, 1

Results: The PMC sets the master to slave mode and then performs a memory test.

Error Indications Test 1 failed **STATUS ERROR
Test 1 failed **VERIFY ERROR

Test 2 -Master Status Test

Command: #MTESTS, J, 2, C

Results: The PMC reads and displays the status of the master on the IMM screen.

LOOP OPEN/CLOSED DEMAND ON/OFF

Error Indications Test 3 failed **INTERRUPT ERROR
Test 3 failed **STATUS ERROR
Test 3 failed **VERIFY ERROR

Test 4 -Message Sequence Verify

Command: #MTEST, J, 4

Results: The PMC sends a sequence of page address and talk data mesages and verifies expected return information.

Error Indications
Test 4 failed **INTERRUPT ERROR
Test 4 failed **STATUS ERROR
Test 4 failed **VERIFY ERROR

Test 5 -Message Sequence Verify

Command: #MTEST, J, 5

Results: The PMC sends a sequence of accept address and accept data messages and verifies expected return information.

Error Indications
Test 5 failed **INTERRUPT ERROR
Test 5 failed **STATUS ERROR
Test 5 failed **VERIFY ERROR

Test 6 -Message Sequence Verify

Command: #MTEST, J, 6

Results: The PMC sends a sequence of talk address and talk data messages and verifies expected return information.

Error Indications
Test 6 failed **INTERRUPT ERROR
Test 6 failed **STATUS ERROR
Test 6 failed **VERIFY ERROR

Test 7 -Message Sequence Verify

Command: #MTEST, J, 7

Results: The PMC sends a sequence of accept test data/talk test data messages and verifies expected return information.

Error Indications
Test 7 failed **INTERRUPT ERROR
Test 7 failed **STATUS ERROR
Test 7 failed **VERIFY ERROR

AUTOMATIC CYCLE MODE

If an intermittent or thermal problem is suspected, it will be necessasry to select one or more of the seven tests to be automatically cycled for long periods of operation. The following command sequence may be used to perform this optional test:

```
#PR, 1G
01 MTEST, J, 1
02 MTEST, J, 6
03 MTEST, J, 7
04 IF, 1, EQ, 1, 1
05 END
```

This example shows tests 1, 6, and 7 selected for continuous cycling. The PMC will continuously run these tests until terminated by the operator with the *BR, CIAD command.

Programs may be composed in the interactive command language and stored internally within CIAD for execution. Such programs are useful for creating continuous message sequences or providing useful predefined diagnostic procedures. Programs may oe entered either from the keyooard or from cassette tape.

PROGRAM, p_1 [,p_2] (CR)

The command shown above is used to create programs for suosequent execution. P_1 specifies the logical unit number of the program input device. p_2 is an optional parameter specifying the program destination device. If omitted (default = 0) the program is read into CIAD's internal program buffer. If p_1 specifies the IMM console device, a prompt is output to the screen giving the line number. The program buffer is always cleared upon execution of the PROGRAM command and the line number is reset to 1. The maximum internal program size is 99. While in the program mode, any of the valid commands (both interactive and programming) may be entered as program steps. If p_3 specifies a cassette drive, the prompt is suppressed and commands are read in from the tape in the same form as the operator entry.

If p_2 is supplied, it should reference the logical unit number of the left or right cassette drive. In this case, the program is not read into the internal buffer but is copied to the cassette tape.

END [,p_1] (CR)

The command shown above is used to terminate the program mode. p_1, if used, specifies the LU device to which the interactive run mode is to transfer for its command input. If defaulted, run control will transfer to the internal program buffer.

FILE, p_1 [,p_2] (CR)

The command shown above is used to select a program or data file from one of several on the cassette tape. p_1 specifies the logical unit number of the drive and p_2 specifies the file number. If p_2 is omitted, the cassette drive will rewind.

LIST (CR)

The command shown above is used to list the current program in the internal program buffer on the IMM console.

DELAY, p_1 [,p_2 [,p_3]] (CR)

The command shown above causes a delay whose length is specified by p_1 in units of 10 microseconds. p_2, if used, specifies a program line number to which control is transferred after the delay. p_3, if used, specifies a loop count. If p_3 is not zero, the delay will loop to line p_2 for p_3 times and then fall through to the next line.

The use of p_2 and p_3 are only legal if the interpreter is running from its internal program buffer.

DELAY commands cannot be nested, i.e., only one DELAY with parameters p_2 and p_3 may be used in a program.

IF,p_1,p_2,p_3,p_4 (CR)

The command shown above is used to make conditional program branches. p_1 and p_3 specify two global variables whose values are compared by the relation given by p_2. If the relational expression is true, control is transferred to the line number given by p_4. If it is false, control falls through to the next line.

Legal entries for p_2 are:

EQ - equal to
NE - not equal to
LT - less than
GT - greater than

The use of the IF command is only legal if the interpreter is running from its internal program buffer.

RUN [,p_1] (CR)

The command shown above is used to change the device from which the interpreter obtains its commands. The interactive command mode is specified when p_1 is the logical unit number of the console. If p_1 is defaulted, run control transfers to the internal program buffer. If p_1 is the logical unit number of the left or right cassette drive, run control passes to the program stored on cassette tape. When running from tape, the "#" prompt is suppressed.

ERROR RECOVERY

1. Care should be taken that LU numbers given in command strings are correctly given. Most errors resulting from wrong LU numbers, may be recovered by using the RTE command:

*OF,CIAD,1

and then restarting with *RU,CIAD

If the equipment table was set "down" as a result of the error, the table should be set "up" again before proceeding. (See Example 1)

2. If CIAD is given a program which causes an infinite loop, the interactive command prompt "#" can be regained by entering the RTE command.

*BR,CIAD

Under certain conditions, a message may never complete if the master fails to interrupt the computer. This looks like an infinite loop but is not. Turn CIAD off (OF, CIAD, 1) and correct the master problem.

Example 1:

You want to send a message to station 3 on channel J to verify that the J-master can communicate with station 3. You do not know which are valid input or output addresses at the station. Use the TLINK command to communicate with the slave station test register, thus avoiding the need to commit himself to a read or write from/to a specific point address:

#TLINK, J, 5, FFFF, IMM You mistakenly entered 5 as station number.

I/O PE L21 E04 501	Station 5 exists on this system but it is not responding to the master so a link error is generated and an RTE system message is generated.
*OF, CIAD, 1	Turns off CIAD
*UP, 4	Sets the J link back into service by using the RTE "UP" command to equipment table 4 (equipment table number obtained from the diagnostic error message).
*RU, CIAD	
#TLINK, J, 3 FFFF, IMM	User enters command correctly
FFFF	CIAD prints result returned
#	Ready for next command

Example 2:

You want to read the state of a logic signal input wired to the CSH point address /JF41/2.

#SET, FG, 2	You select HEX formatting so that the IMM parameter in the following commands will result in data being displayed in HEX form.
#RLINK, J, FH, 41H, IMM	Read/JF41/ and display the contents on the IMM.
3A04	The HEX content of /JF42/ is displayed. The binary equivalent of 3A04 is 0011101000000100 and bit 2 is seen to be true.
#	Ready for next command.

Example 3:

You want to verify that an integer count variable is being transmitted to the slave K channel of this PMC from another PMC at point address 60H. You have learned, using the IMM functions that the current value of the signal of interest is 395 (decimal).

#SET, FG, 1	You select Decimal formatting.
#RLINK, K, 0, 60H, IMM	You read/K060/ and display the contents on the IMM.
395	The decimal value is displayed.

#	Ready for next command.

Example 4:

You want to check the operation and calibration of an NDAC digital-to-analog converter card at link address /JA05/. The format of the 12-bit NDAC is such that the left-most 12 bits are significant and -10 volts corresponds to an output of 8000H. 0 volts corresponds to an output of 0000H, and +10 volts corresponds to 7FF0H.

#SET, FG, 2	You select HEX formatting.
#PROGRAM, IG 01 WLINK, J, AH, 5, IMM 02 DELAY, 100, 1, 2 03 END	You create a 2-line program which will write the value entered from the IMM to the NDAC card and then loop continuously with a 1-second delay for a total of 3 values.
0000	Enter the first value to be output. Then check and adjust the NDAC if necessary.
8000	Enter the second value to be checked.
7FF0	Enter the third value.
#	Ready for next command.

Example 5:

You want to write a program and save it on cartridge tape to be used to exercise all slave stations on a CSH loop. In this example, channel J has stations 1, 2, 3 and F as valid stations. The program will send a test message to each station in turn and verify that the value read from the test register agrees with the value written. The program will continuously cycle until an error occurs when it will pause and display the Faulty Station number on the IMM.

Place a blank tape in the left drive and rewind.

#PROGRAM, IG, LG	Write program onto left tape
01 SET, 0G, 0	Constant for test register = 0
02 SET, 1G, 1	Select station 1
03 TLINK, J, 1G, 2G	Write from 0G and read to 2G
04 IF, 0G, NE, 2G, 16	Compare - exit if not identical
05 SET, 1G, 2	

```
06 TLINK, J, 1G, 0G, 2G          Repeat for station 2
07 IF, 0G, NE, 2G, 16
08 SET, 1G, 3
09 TLINK, J, 1G, 0G, 2G          Repeat for station 3
10 IF, 0G, NE, 2G, 16
11 SET, 1G, FH
12 TLINK, J, 1G, 0G, 2G          Repeat for station F
13 IF, 0G, NE, 2G, 16
14 CA, 0G, +, 1                  Increment test register constant
15 DELAY, 1, 2                   Loop 100 times/second to line 2
16 **TEST REGISTER ERROR ON STATION
17 SET, IMM, 1G                  Output error message
18 END, 1G                       End program, transfer control to IMM
#FILE, LG, 1                     Find program File (File #1)
#PROGRAM, LG                     Read program to internal buffer
#RUN                             Run diagnostic until error
*BR, CIAD                        Operator interrupt of CIAD
#ABORT                           End user session
```

HIGHWAY TEST MODULE

The Highway Test Module (HTM) is a special purpose troubleshooting facility designed specifically for testing the CSH. The HTM is described in detail in GEK-65151. It is not a substitute for CIAD, but is an optional adjunct to it. We highly recommend the Highway Test Module as a very effective item of test equipment.

Master Mode Operation:

* Works in conjunction with the MTEST commands of CIAD only.
* Passive to WRITE/COMPARE of CSH local Ram by PMC (MTEST 1)
* HTM can simulate I/O demands and cable breaks (MTEST 2)
* Can respond to PMC as to validity of its directorship of message activity (MTEST 3 thru 7)

Slave Mode Operation:

* Can originate messages at head of loop (Tests 1 through 12).
* Can check validity of data and address on I/O bus of a slave station (Tests 13 through 19). (See Figure 6-1).
* Can simulate missing I/O bus hand shake signals (Test 24). (See Figure 6-1).
* Can write/compare CSH local Ram, (Tests 20 through 22). (See Figure 6-2).

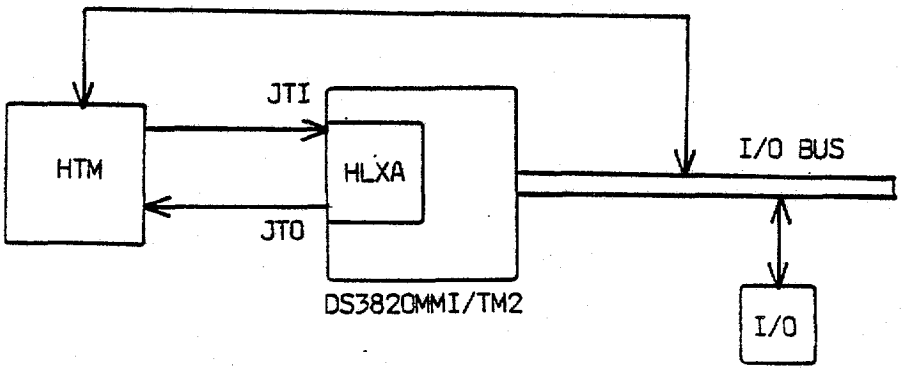


Figure 6-1.
I/O Test Setup

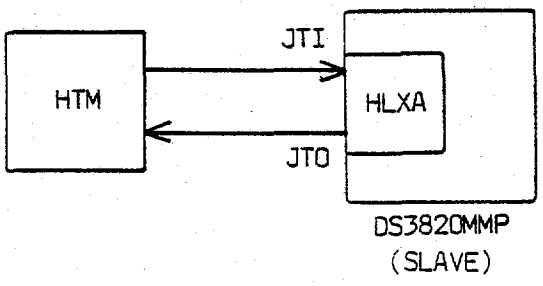


Figure 6-2.
Master Station in Slave Mode
Test Setup

Input/Output Operation

* Connected to by-passed slave station (simulate activity). Can transmit to or receive from selectable I/O. (See Figure 6-3).

* Connected to loop (monitor activity). (See Figure 6-4). Can monitor message stream for selectable data-supplying or data-gathering message.

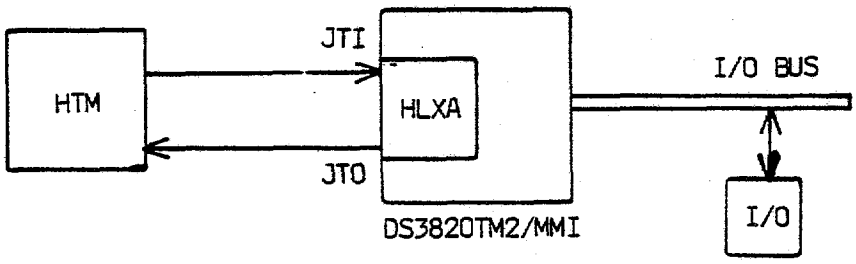


Figure 6-3.
Slave Station Test Setup

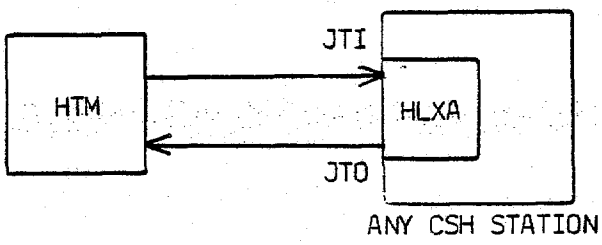


Figure 6-4.
CSH Monitoring Configuration

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